LOCAL FOOD DISTRIBUTION SYSTEM FOR HUNGER RELIEF

By

SEHRISH SIDDIQUE

Supervised By

Dr. Akmal Saeed Khattak

Department of Computer Science

Quaid-i-Azam University Islamabad

(2022-2026)

CERTIFICATE OF APPROVAL

Supervisor: ______________

Dr. Akmal Saeed Khattak

Department of Computer Science

Quaid-e-Azam University Islamabad,

Chairperson: _______________

Prof. Dr. Ayyaz Hussain

Department of Computer Science

Quaid-e-Azam University Islamabad,

Date: _____________

ACKNOWLEDGEMENT

All praises and thanks are to Allah Almighty the most Gracious and the most Merciful, who gave me enough strength, patience and guidance to complete this project and my Academic journey at Quaid-e-Azam University.

I would like to thank and show my gratitude to Sir Akmal Saeed Khattak, for his continuous guidance, his valuable suggestions and beneficial feedback and his continuous help in this project which helped me not only complete this project but also have deep understanding of the project.

I would also like to thank the faculty of Computer Science Department, including Dr. Rabeeh Ayaz Abbasi, Dr. Khalid Saleem, Dr. Muhammad Shuaib Karim, Dr. Onaiza Maqbool, Dr. Muazzam Ali Khan Khattak, Dr. Ayyaz Hussain, Dr. Mudassar Azam Sindhu, Dr. Ghazanfar Farooq Siddique, Dr. Umer Rashid, and Mam Ifrah Farukh Khan, for providing all the knowledge in my academic journey, their guidance, and motivation helped me throughout the degree.

Lastly, I would like to thank my parents for their continuous support and guidance that helped me in challenging times.

ABSTRACT

Food wastage and food insecurity that exist in the world as a social problem and a huge challenge only due to the lack of coordination. Donors find it difficult to find right NGOs at time to donate food. And there is also the lack of trust in NGOs and Donors.

Focus of this project is the development of web based Food Distribution System designed to digitalize the process of donating food that is in excess in households or restaurants to reduce food wastage and facilitate timely donations.

This system uses the MERN stack to ensure efficient experience. The system allows the donors to donate excess food and NGOs to accept that food and distribute it to the people. The primary objective of the project is to make reliable, efficient and organized charity management systems that smooth the user experience.

Revision History

Version No.	Date	Modified By	Significant Changes
1	1/AUG/2026	Sehrish	Original
2	2/AUG/2026	Sehrish	Updated ERD & System Workflow
2	15/AUG/2026	Sehrish	Added Software Implementation & Conclusion
2	21/AUG/2026	Sehrish	Added Test Results & Changed STD to Chapter 5

Table of Contents

Software Test Document (STD)

Conclusion

List of Figures

Software Project Management Plan (SPMP)

List of Tables

Software Project Management Plan (SPMP)

1. Software Project Management Plan (SPMP)

1.1 Introduction

Poverty and hunger are still one of the world's main problems that still exists in the world despite all the development that man has made and all the problems the man has solved. A significant population of the world still suffers from hunger and are unable to meet basic daily food.

Figure 1.1 [1]

In our everyday life, a lot of food is wasted intentionally and unintentionally. In our households, where there is not large number of leftovers, they are wasted not because they intentionally want to waste them but because they can't find any volunteer at perfect time to give their food before the expiry time of the food same in restaurants and big places find too many needy people because it is also difficult.

Figure 1.2 [2]

And on other hand there are a lot of people who cannot even afford daily basic food, they are approached by NGOs. But again, many of the donors are unable to communicate to NGO at perfect time, where they want to donate their food before the expiration of the food. And even donors find one there is too much hustle of donating the food at location, there comes a lot of trust issues between the donors and the receivers, there is delivery hustle whether the donation has arrived at destination, communication gaps who will deliver the donation which makes donation difficult.

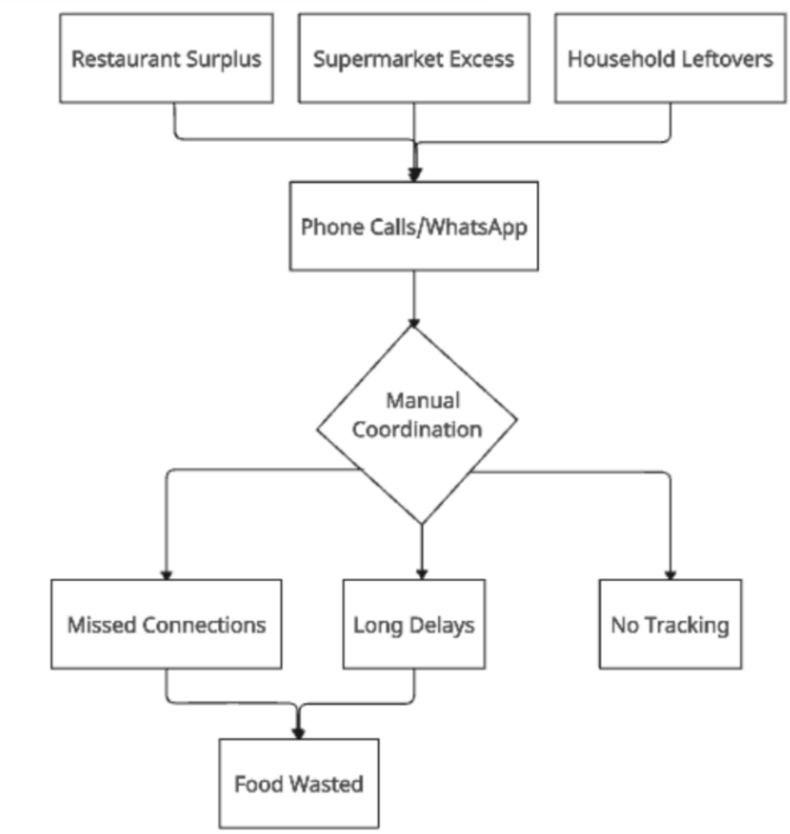


Figure 1.3: Manual coordination System Workflow Diagram

Figure 1.3 shows the flow of normal donation that takes Place, how coordination between donors and receivers happens, resulting in non-effective experiences. It shows in manual coordination the donor donate through phone calls and result in missed connections, long delays and lack of tracking.

1.1.1 Project Overview

The Local Food Distribution System for Hunger Relief is a web based technology solution that is designed to connect the donors that want to donate surplus food with the NGOs who want to accept that food to help them distribute it to people in need managed by the administration that ensures the safe coordination between donors and NGOs. The platform smooths communication between donors by website effective features.

1.1.2 Motivation

The project aims to enable smooth communication between donors and NGOs, tells the status of delivered donations whether they have reached or not, ensures proper entrusted and authentic receivers of donation and most importantly enables finding receivers as soon as possible by giving many receivers to join the platform and alerting them of the donations.

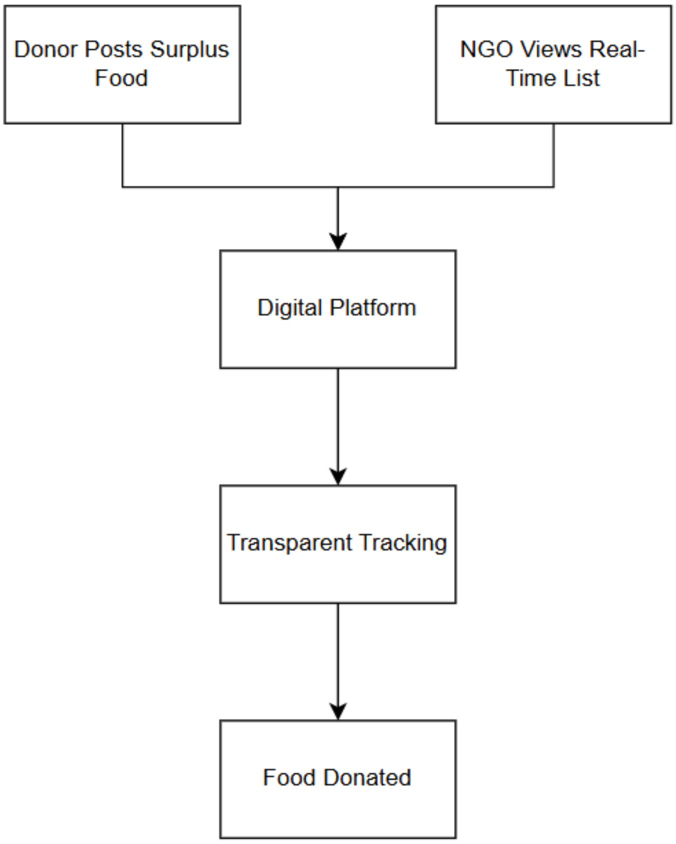


Figure 1.4: Proposed System Workflow Diagram

Figure 1.4 shows the flow of proposed donation. It shows that there will be streamline and managed process containing transparent tracking through digital system which results in successful food donation.

1.1.3 Existing Systems

2. **Specific NGO-based Platforms**

Example: Edhi Foundation application

Source: https://www.edhi.org/donate-us

Limitations: They enable food donation but big drawback in them is there you can only donate money not actual food and if there are some websites that directly donate food, it has only one receiver which makes the less utilization of all donations and over burdens that specific receiver

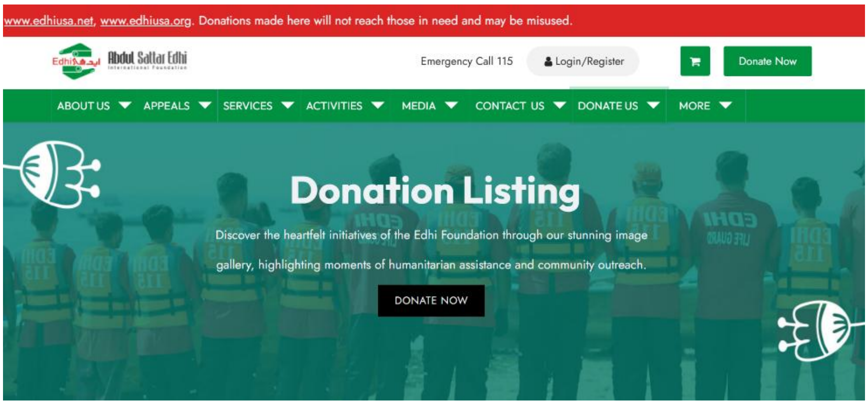


Figure 1.5: Eidhi foundation homepage

Figure 1.5 shows homepage of Eidhi Foundation. It basically shows we can donate to it but only to one NGO.

3. Multi NGOs (receivers) based Platforms

Example: Food Rescue US application

Source: https://foodrescue.us/

Limitations: They have multiple receivers and donors that donate food, they can view available food ,collect them but big missing thing in these applications is that there is no feature donation status tracking for user to check current status of the donation.

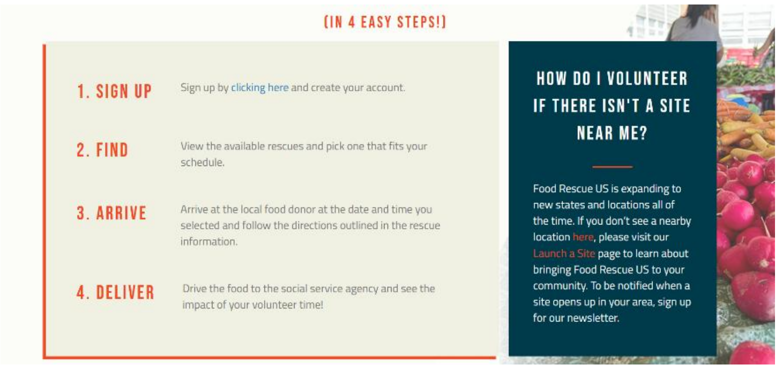


Figure 1.6: Food Rescue Us Guidelines

Figure 1.6 shows the steps of donation at Food Rescue Us Guidelines. It shows the people have to deliver from donor to receiver by signing up then finding the donor and then collecting then delivering it.

4. **Social media-based groups**

Example: WhatsApp, Facebook etc.

Limitations: Groups form on these platforms enable the donation, but it causes too much hustle and not any clarification of donations volunteers and donors.

5. **Traditional Manual Platforms**

The traditional donation way based on emails and phone numbers. Which basically is very rare to work now as they don't provide good communication.

1.1.4 Project Deliverables

1. Web Application

One of the Project Deliverables is Web Application that is **react js** based. It provides an interface for donors to post donations and keep track of their donations, NGOs to accept donations and be updated with the latest donations and lastly Admin to verify the NGOs and manage the platform.

2. Backend System

The project also delivers backend system made with **Node.js/Express** providing

authentication and authorization, with RESTful API for all operations to be carried on the platform, to make data models and schemas and building notification systems.

3. Database

Database made by **MongoDB** delivers donors, NGOS and admin management tables and tracking of all the donations and their status and all other information required to store of all users, platform and donations.

4. Documentation

Documentation provides the project plan, all requirements, use cases and other things with user manuals of all users with technical documentation for developers, and guide for system administration so that application can be made effectively.

1.2 PROJECT ORGANIZATION

1.2.1 Software Process Model

The project uses Agile Framework to ensure small, fast and quick implementation of the steps to get feedback from the stakeholders on time and evolve the project effectively in sprints, it is also useful as it ensures timely meeting with stakeholders to get feedback and give time to conduct iterative testing , enabling less documentation and more working on the software.

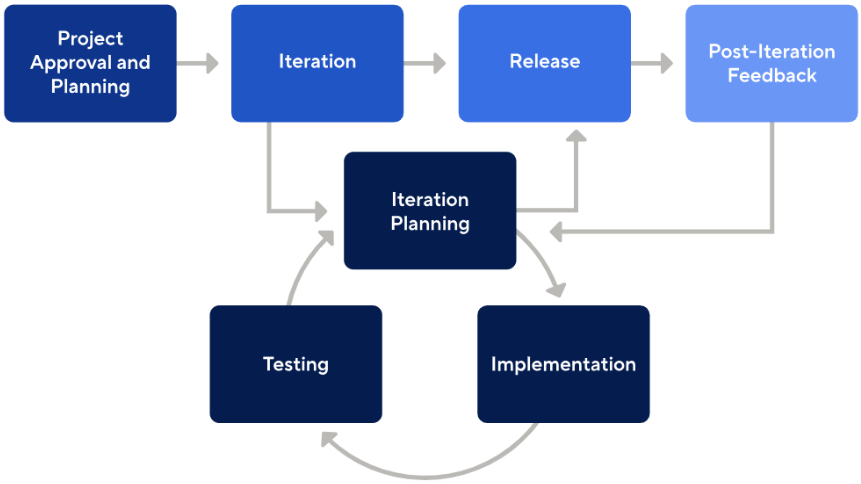


Figure 1.7: Agile Workflow Diagram [3]

Figure 1.7 shows the Agile Workflow Diagram. It shows that the project will be done in iterations and for each iteration it will be released after its planning, implementation and testing.

1.2.2 Roles and responsibilities

Project Manager (Sehrish Siddique)

Here, Project Manager plays the role of not only manager but also Frontend Developer, Backend Developer, Database Administrator, UI/UX designer and quality assurance engineer.

As of all, Project Manager in Frontend handles responsive, smooth easy to understand interfaces, in Backend Project manager handles all the RESTful APIs, data model and schemas with authentication and authorization and notification system, in database handles the database design and performance optimization, in Quality assurance handles and test making and their execution.

Stakeholder (Sir Akmal Khattak)

Stakeholders join all weekly meetings and give feedback. Stakeholders specify the requirements and monitor the progress of software development with test conduction.

1.2.3 Tools and techniques

Frontend Development

- React js
- Javascript, css, html
- Vs code

Backend Development

- Node js/Express
- JWT
- postman
- vs code

Database

- MongoDB /MongoDB Atlas

Documentation

- MS word
- Draw.io
- SPMP
- SRS
- SDD
- STD

1.3 PROJECT MANAGEMENT PLAN

1.3.1 Tasks

1. Requirements
2. Analysis
3. Design
4. Implementation
5. Testing

1.3.2 Description

1. **Requirements**

In this task, all the important requirements are gathered, whether it involves what is physically needed for building the project or in what manner execution of project will go on and who will be stakeholders etc.

2. **Analysis**

In this task, analyzation of the requirements takes place, what would be the logic, use cases and flows that need for building and will be needed in design phase.

3. **Design**

Here, the actual technical assessment starts, defining system architecture, data structures and designing interfaces takes place here

4. Implementation

All the planning made in previous phases are implemented in this phase. This is where the coding starts and the project is built.

5. Testing

Here, the built project tests whether the things built are either working or not. It tests whether the components built are reliable and functional.

1.3.3 Schedule Summary

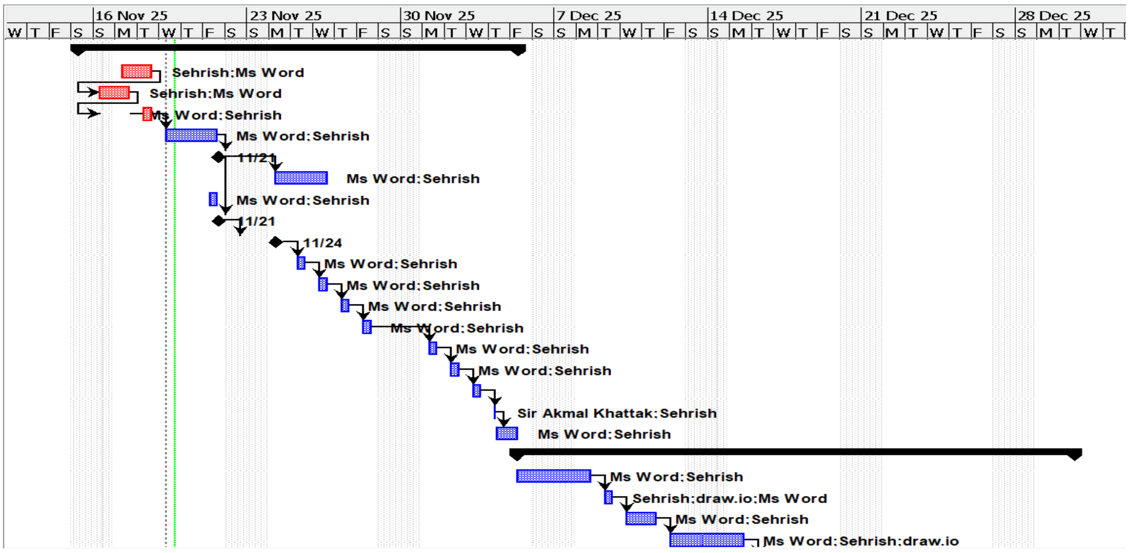
1. 1 week for gathering requirements
2. 2 weeks for analysis
3. 3 weeks for design
4. 5 weeks for implementation
5. 1 week for testing

1.4 Timetable

1.4.1 Project Plan

Work Activities

The following table shows all the tasks and activities to be performed, it basically gives an overview in a flow to whom the tasks are appointed, the resources and the tasks and after the activities what activities will carry out.



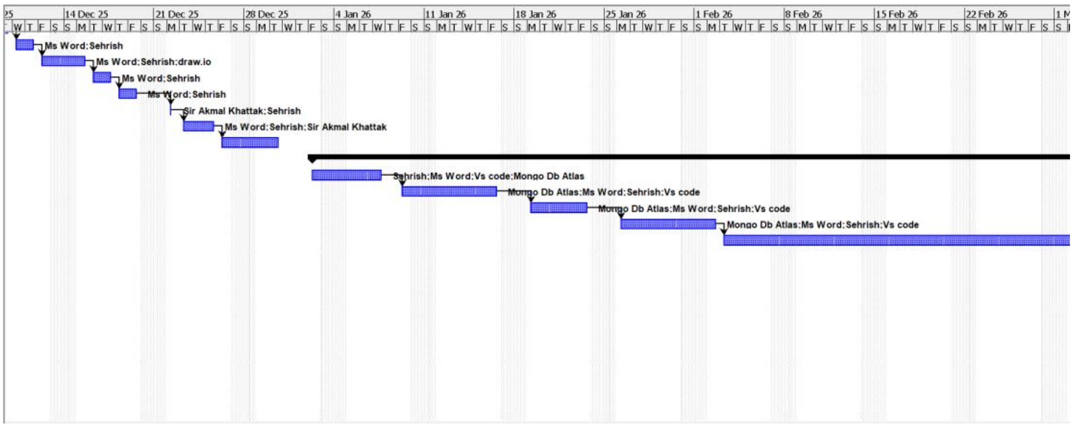


Figure 1.8: Work Activities Diagram

Figure 1.8 gives the visualization of all work activities that after the particular activity with its actors which activity will be performed , it gives it visualization.

Resource Allocation

The following table shows the list of all activities and their schedule allocations , with their starting and ending dates goals, what will be total required days for them.

	Name	Duration	Start	Finish	Predecessors	Resource Names
1	**⊟ Planning**	14.25 days	**11/15/25, 8:00 AM**	**12/5/25, 10:00 AM**		
2	Gather requirements	2 days	11/15/25, 8:00 AM	11/18/25, 5:00 PM		Sehrish;Ms Word
3	Develop Project Deliverab	2 days	11/16/25, 8:00 AM	11/17/25, 5:00 PM	2	Sehrish;Ms Word
4	Choose Project Model	2 days	11/16/25, 8:00 AM	11/17/25, 5:00 PM	3	Ms Word;Sehrish
5	Define Responsibilities	3 days	11/19/25, 8:00 AM	11/21/25, 5:00 PM	4	Ms Word;Sehrish
6	Define Tools& techniques	0 days	11/21/25, 5:00 PM	11/21/25, 5:00 PM	5	Ms Word;Sehrish
7	Develop Project Plan	3 days	11/24/25, 8:00 AM	11/26/25, 5:00 PM	6	Ms Word;Sehrish
8	Define Scope	1 day	11/21/25, 8:00 AM	11/21/25, 5:00 PM		Ms Word;Sehrish
9	Define Objectives	0 days	11/21/25, 5:00 PM	11/21/25, 5:00 PM	6	Ms Word;Sehrish
10	Define Constraints	1 day	11/22/25, 5:00 PM	11/24/25, 5:00 PM	9	Ms Word;Sehrish
11	define Assumptions	1 day	11/25/25, 8:00 AM	11/25/25, 5:00 PM	10	Ms Word;Sehrish
12	Define Project functions	1 day	11/26/25, 8:00 AM	11/26/25, 5:00 PM	11	Ms Word;Sehrish
13	Define hardware reuireme	1 day	11/27/25, 8:00 AM	11/27/25, 5:00 PM	12	Ms Word;Sehrish
14	Define Software Requirem	1 day	11/28/25, 8:00 AM	11/28/25, 5:00 PM	13	Ms Word;Sehrish
15	Define Functional requirer	1 day	12/1/25, 8:00 AM	12/1/25, 5:00 PM	14	Ms Word;Sehrish
16	Develop SPMP	1 day	12/2/25, 8:00 AM	12/2/25, 5:00 PM	15	Ms Word;Sehrish
17	Milestone	1 day	12/3/25, 8:00 AM	12/3/25, 5:00 PM	16	
18	Review SPMP	0.25 days	12/4/25, 8:00 AM	12/4/25, 10:00 AM	17	Sir Akmal Khattak;Sehrish
19	Refine SPMP	1 day	12/4/25, 10:00 AM	12/5/25, 10:00 AM	18	Ms Word;Sehrish
20	**⊟ Analysis**	18 days	**12/5/25, 8:00 AM**	**12/30/25, 5:00 PM**		
21	Define Usecases	2 days	12/5/25, 8:00 AM	12/8/25, 5:00 PM		Ms Word;Sehrish
22	Develop UC Diagram	1 day	12/9/25, 8:00 AM	12/9/25, 5:00 PM	21	Sehrish;draw.io;Ms Word
23	Write UseCase Descriptio	2 days	12/10/25, 8:00 AM	12/11/25, 5:00 PM	22	Ms Word;Sehrish
24	Develop SSDs	2 days	12/12/25, 8:00 AM	12/15/25, 5:00 PM	23	Ms Word;Sehrish;draw.io

		Name	Duration	Start	Finish	Predecessors	Resource Names
25		Define Domain Model	2 days	12/16/25, 8:00 AM	12/17/25, 5:00 PM	24	Ms Word;Sehrish
26		Define Database Requirer	2 days	12/18/25, 8:00 AM	12/19/25, 5:00 PM	25	Ms Word;Sehrish
27		Develop SRS	0.5 days	12/22/25, 8:00 AM	12/22/25, 1:00 PM	26	Sir Akmal Khattak
28		Review SRS	3 days	12/23/25, 8:00 AM	12/25/25, 5:00 PM	27	Ms Word;Sehrish;Sir Akmal ...
29		Refine SRS	3 days	12/26/25, 8:00 AM	12/30/25, 5:00 PM	28	
30		**⊟ Design**	**55 days**	**1/2/26, 8:00 AM**	**3/19/26, 5:00 PM**		**Sehrish**
31		Develop System Architect	4 days	1/2/26, 8:00 AM	1/7/26, 5:00 PM		Sehrish;Ms Word;Vs code;...
32		Develop SDs	6 days	1/9/26, 8:00 AM	1/16/26, 5:00 PM	31	Mongo Db Atlas;Ms Word;S...
33		Develop EERD	5 days	1/19/26, 8:00 AM	1/23/26, 5:00 PM	32	Mongo Db Atlas;Ms Word;S...
34		Design UI Screens Prototy	6 days	1/26/26, 8:00 AM	2/2/26, 5:00 PM	33	Mongo Db Atlas;Ms Word;S...
35		Develop SSD	21 days	2/3/26, 8:00 AM	3/3/26, 5:00 PM	34	Mongo Db Atlas;Ms Word;S...
36		Review SDD	1.5 days	3/4/26, 8:00 AM	3/5/26, 1:00 PM	35	Sir Akmal Khattak;Sehrish
37		Refine SDD	9 days	3/9/26, 8:00 AM	3/19/26, 5:00 PM	36	Mongo Db Atlas;Ms Word;S...
38		**⊟ Testing**	**6 days**	**3/21/26, 8:00 AM**	**3/30/26, 5:00 PM**		
39		Define Testing Approach	0 days	3/21/26, 8:00 AM	3/23/26, 5:00 PM		Sehrish
40		Define Test Plan	1 day	3/24/26, 8:00 AM	3/24/26, 5:00 PM	39	Sehrish
41		Write Test Cases	1 day	3/25/26, 8:00 AM	3/25/26, 5:00 PM	40	Sehrish
42		Develop STD	1 day	3/26/26, 8:00 AM	3/26/26, 5:00 PM	41	Sehrish
43		Review STD	1 day	3/27/26, 8:00 AM	3/27/26, 5:00 PM	42	Sir Akmal Khattak
44		Refine STD	1 day	3/30/26, 8:00 AM	3/30/26, 5:00 PM	43	Sehrish

Table 1.1: Work Activities Table

Table 1.1 shows All the tasks and process to be performed within their timeline when the activity will start when will it end and what will be its resources .

Resources Needed

The image shown below is basically estimation of the list of all the resources that are needed to make the project and what is the type of that resource.

		Name	RBS	Type	E-mail Address	Material Label	Initials
1		Sehrish		Work			S
2		Sir Akmal Khattak		Work			S
3		Ms Word		Material			M
4		Mongo Db Atlas		Material			M
5		Vs code		Material			V
6		draw.io		Material			d

Table 1.2: Resources Needed Table

Table 1.2 shows all the resources to be used in the project, what are the resources their types and initials.

2. Software Requirements Specification (SRS)

2.1 Introduction

Software requirements specifications describe all the requirements for the platform **Local Food Distribution System for Hunger Relief**. It describes the scope of the system, its constraints, all the assumptions that will be made making this system and most importantly all the functional and non-functional requirements of the system. This chapter also describes all the use cases of the system with description for enhanced clarity.

2.1.1 Scope

The scope of **Local Food Distribution System for Hunger Relief** is the development of the system that makes donors and NGOs donate and accept donations with proper security in minimum time. It has three modules: Donor, NGO, and Admin. It involves real time donations tracking with their current status, NGO documents verification and NGO requesting admin to accept those donations. And Admin handling all donations and donors, NGO communication.

2.1.2 Objectives

1. Developing a platform that contributes to minimizing food wastage.
2. To enable NGOs to be updated with latest donations.
3. Allowing the Admin to monitor the system.
4. Marinating a system that ensures transparency of donations.
5. Building a trusted platform for donations.

2.1.3 System Workflow

Figure 2.2 shows the workflow of the system, how the thing works in the system, the figure depicts that at first the donor will register itself in order to post donation and then logs into the system and post the donation to donate the surplus food and then the donation is posted with the status available and in order to accepts those donations NGO registers into the system and admin approves or rejects the NGO , if the NGO is rejected by the admin then NGO cannot log into the system and accept donations but if the NGO is approved by the admin then NGO can log into the system and view all the available donations viewed by the donors and request any available donation then the donation status for that donation becomes requested with request status pending and then admin approves or reject the request of the NGO , if the donation request of NGO is approved then the request status becomes pending then that donation is booked for that NGO and the status of the donation becomes booked and then the NGO picks up the donation from pickup location of donor and marks the donation status as collected but if the donation is rejected, then NGO cannot collect it and its request status becomes rejected.

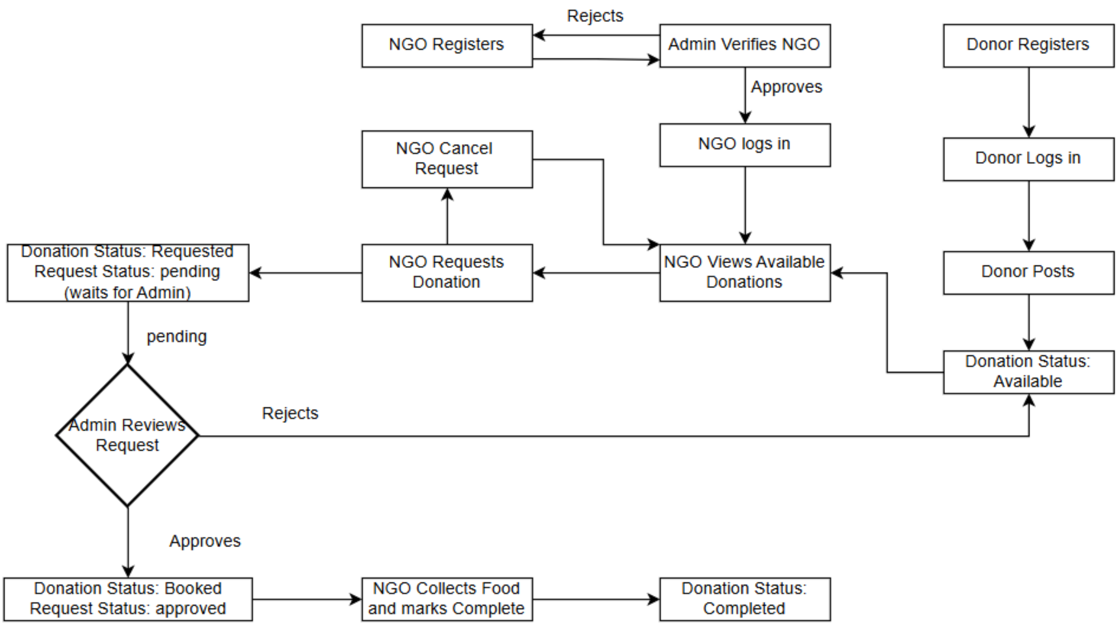


Figure 2.1: System Workflow Flowchart

2.1.4 Constraints

1. Platform must work on chrome.
2. Platform must be free.
3. Ngo cannot request to book donation until NGO is verified by the administration.
4. Ngo can only accept 5 donations per day.
5. Donors cannot accept donations.
6. NGO cannot post donations.
7. Admin cannot post or accept a donation.

2.1.5 Assumptions

1. Users have stable internet.
2. Donors will provide accurate information like location, number and donation information.
3. NGO will honestly represent their operational capacity.
4. Admin will honestly perform its duties like verification.
5. Admin will allot donations unbiasedly.

2.2 Overall Description

2.2.1 Product Perspective

Local Food Distribution System is a standalone system, that operates for donations to execute seamlessly by connecting donors, NGOs and admin.

Relationship with external are below:

1. **Cloud Storage:**
   Cloud Storage is used for NGOs documents. This external system is joined by the system to upload the documents of NGOs. All the documents uploaded will be stored on cloud storage.

2.2.2 User Characteristics

1. Donors

- Technical proficiency is basic to high.
- Can register, post donations and track them easily.

2. NGOs

- Technical proficiency is medium to high.
- Can upload documents, accept the donation and track history.

3. Admin

- Technical proficiency is high.
- Can understand system operation and data management.
- Responsible for decision making.

2.3 Functional Requirements

Following are the functional requirements:

Authentication and User Management

1. Users can sign up, login and logout.
2. Can view the notification.
3. System implements JWT-based authentication with session management.
4. System shall implement role based access control.

Donor

5. Can post new donations.

6. Can view list of all its donations and their status whether available, requested, booked or collected.
7. Can view and manage their profile.
8. Can edit and delete available donations.

NGOs

9. NGOs can post documents for verification.
10. Can view all the donations posted.
11. You can send book requests to a donation for pickup and wait for the approval from admin to book it.
12. Mark donation as collected after pickup.
13. Can view and manage their profile.
14. Can view list of all its donations booked and collected.
15. Can view list of all its requested donations.

Admin

16. Verify the NGOs signup.
17. Admin can delete users (donors and NGOs).
18. Can view users (donors and NGOs).
19. Can view all donations based on status (available, booked, collected)
20. View multiple NGOs book requests per donation from NGOs
21. Accept the donation book request and all other donations are auto rejected.
22. Can change the status of the donation.
23. Can delete expired donations.

General

24. Auto deleted expired donations.
25. System allows multiple NGOs to request same donation.
26. System auto-reject other NGOs when admin approves one NGO request.

2.4 Non-Functional Requirements

Reliability and Availability

- Data must be backed up daily automatically so that no loss occurs.
- Notification system should ensure that 99% sent notifications sent are delivered successfully.

Security

- Users must be authenticated by using login credentials.
- Each rule must be enforced strictly and must not be surpassed by other users.Sensitive data like passwords and other information must be encrypted.

Usability

- The interface should be easy for users to interact with.
- Notifications should be clear.

Portability

- The platform should be accessible from Chromes as it should be web-based.
- It should not be operating system dependent.

2.5 Use Cases

2.5.1 Use Cases List

General

UC-1: Login

UC-2: Logout

UC-3: View notifications

Donor

UC-4: Donor Signup

UC-5: Post Donation

UC-6: Edit Available Donation

UC-7: Delete Available Donation

UC-8: View Donor Donation history

UC-9: View Donor Profile

UC-10: Manage Donor Profile

NGOs

UC-11: NGO Signup

UC-12: View Available Donations

UC-13: View request History (Pending, Approved and Reject)

UC-14: Request Available Donation

UC-15: Mark Booked Donations as Collected

UC-16: View NGO Donation history

UC-17: View NGO Profile

UC-18: Manage NGO Profile

Admin

UC-19: Verify NGOs Signup

UC-20: Delete User

UC-21: View User List

UC-22: View all donations

UC-23: View NGO requests

UC-24: Accept requested donation

UC-25: Manage Donation Status

2.5.2 Use Cases Diagram

Figure 2.2 is the use case diagram of the project , the project is divided into three modules, donor, NGO and Admin. Software Construction allows us to divide main use case diagram in parts for the clarity and to understand the system more effectively, that's why in the following rather than showing all the use cases at once and making a mess, usecase diagram is divided in respective parts showing single usecase diagram. Donors use cases are Donor Signup, Post Donation, Edit Available Donation, Delete Available Donation, View Donor Donation history, View Donor Profile, Manage Donor Profile, for NGO are NGO Signup, View Available Donations, View request History, Request Available Donation, Mark Booked Donations as Collected, View NGO Donation history, View NGO Profile, Manage NGO Profile and for admin are Verify NGOs Signup, Delete User, View User List, View all donations, View NGO requests, Accept requested donation**,** Manage Donation Status and for all user are Login, Logout, View notifications

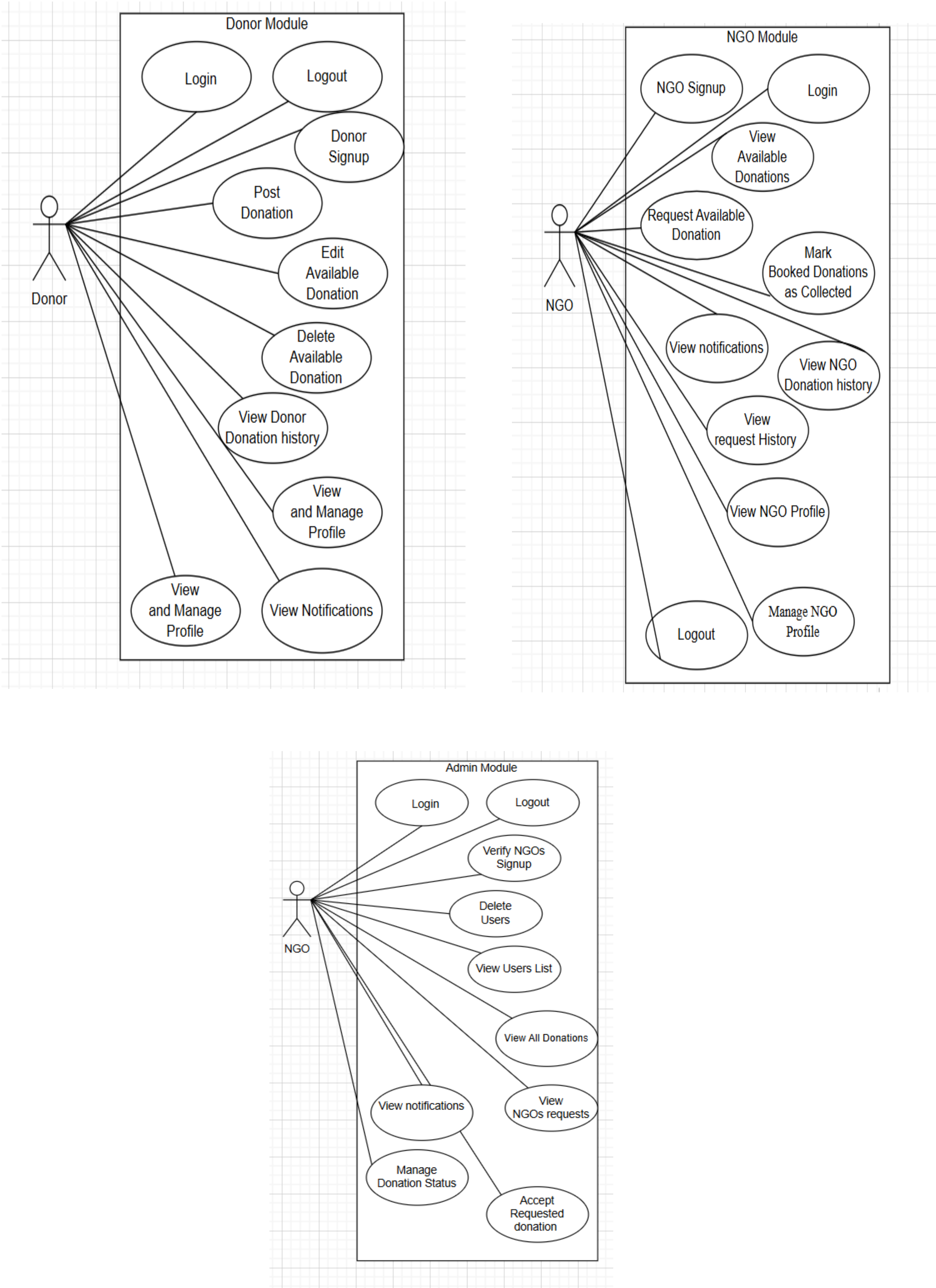


Figure 2.2: Usecase diagram

2.5.3 Use Cases Description

UC1. Login

Table 2.1 shows the usecase description of login for the user that are admin ,NGO and donor selects login option with inputs email and password to the system, system process the validation of information entered by users in database and if the user exists the user is then redirected to respective user module and if user adds invalid information, it gives error message of invalid credentials.

ID	1
Name	Login
Primary Actor	Donor, NGO, Admin
Stakeholders and Interests	**Users:** for Donors, NGOs and Admin to have access to the platform to do the tasks they need to do. **Security Team:** to ensure that only authenticated and existing users have access to the platform.
Preconditions	1. Users have already signed up. 2. System is working.
Postconditions	1. Users are logged in and have access to the platform of their specific module. 2. Session of that user is settled as active.
Main Success Scenario	1. User selects the login button. 2. User is redirected to login form. 3. User adds login credentials to the form. 4. User selects submit option. 5. System checks the validation of information entered by users in database. 6. System generates the JWT token. 7. System redirects user to the respective page according to the user.
Extensions/Alternative Flows	**3A** Invalid login credentials 1. System gives the message of invalid credentials. 2. Users retry. **3B** NGO account not verified 1. System gives the message of unverified account.
Frequency of Occurrence	High
Inputs	1. Email 2. Password
Outputs	1. Access to user specific page 2. Authentication token

Table 2.1: Login

UC2. Logout

Table 2.2 shows the usecase description of logout for users Admin, Ngo and donor. It show the flow of logout that user selects logout button and system invalidates its authentication and redirects the user to login page and if the session timeout occurs even before the logout then system displays the session timeout message and redirects the user to login page.

ID	2
Name	Logout
Primary Actor	Donor, NGO, Admin
Stakeholders and Interests	**User**: want to close the session **Security team:** to ensure termination of session securely
Preconditions	1. User is logged in.
Postconditions	1. Session is ended securely. 2. User is redirected to login page.
Main Success Scenario	1. User selects the logout button. 2. System invalidates the authentication token. 3. System ends the active session. 4. System redirects users to login page.
Extensions/Alternative Flows	1A Session timeout 1. System logout before the user requests to log out due to timeout. 2. System gives messages of session timeout. 3. System redirects users to login page.
Frequency of Occurrence	High
Inputs	1. Logout request
Output	1. Session terminated

Table 2.2: Logout

UC3. View Notifications

Table 2.3 shows the usecase description of view notifications, user chooses view notifications option, and the system displays the list of all the notifications by organizing them chronologically and if no notification in the system, then the system displays the message of no notifications to view.

ID	\
Name	View Notification
Primary Actor	Donor, NGO, Admin
Stakeholders and Interests	Donor: want to view all confirmation notifications and donation collection messages. NGO: want to view all confirmation notifications. Admin: want to view all confirmation notifications with verification messages.
Preconditions	1. User has logged in.
Postconditions	1. User can view list of notifications.
Main Success Scenario	1. User chooses ‘Notification’ option. 2. System retrieves all the notification of that user. 3. System organizes them according to date. 4. System displays the notifications. 5. User can view the notifications.
Extensions/Alternative Flows	2A. No notification 1. System displays the message of no notifications
Frequency of Occurrence	High
Inputs	1. User id
Outputs	1. List of notifications

Table 2.3: View Notifications

UC4. Donor Signup

Table 2.4 shows the usecase description of donor signup, donor chooses signup option with details that are name, email, password, contact number and location, then the system redirects the user to login page and if the signup information provided by user is incomplete , then the system displays the message of incomplete information and if the signup email provided by user is already exists, then the system displays the message of it.

ID	4
Name	Donor Signup
Primary Actor	Donor
Stakeholders and Interests	**Donor:** wants to quickly sign up to post donation. **Admin:** Wants to get accurate information of donors to handle platform.

Preconditions	1. User has not already signed up. 2. System is working and connected to database.
Postconditions	1. A new donor added to database. 2. Users can login.
Main Success Scenario	1. User selects the signup option. 2. User is redirected to the signup form. 3. User fills all the information required in signup form. 4. User selects the submit option. 5. System checks the validation of information entered by users. 6. System creates new users. 7. System gives messages of new users created. 8. User is redirected to login page.
Extensions/Alternative Flows	**4A.** Incomplete information 1. System shows the message of incomplete information. 2. User enters the incomplete information. **5A.** Duplicate email. 1. System shows the message that email already exists. 2. User enters the new email. **6A**. System failure 1. System gives the message of system failure. 2. Account is not created.
Frequency of Occurrence	High
Inputs	1. User type 2. Name 3. Email 4. Password 5. Location 6. Contact number
Outputs	1. User created message 2. User id generated 3. User stored in database

Table 2.4: Donor Signup

UC5. Post Donation

Table 2.5 shows the usecase description of post donation, donor chooses post donation option and is rendered to form of posting donation ,donor adds and submit all information required, and the system displays the message of donation posted and if information provided by donor is incomplete, then the system displays the message of invalid detail.

ID	5
Name	Post donation

Primary Actor	Donor
Stakeholders and Interests	**Donor:** wants to post donation. **Admin:** Wants to get accurate information of donor donation to handle platform. **NGO:** wants donations to receive them.
Preconditions	1. Donor has logged in.
Postconditions	1. Donation is posted with available status. 2. NGOs receive donation notification. 3. Ngo and admin can view.
Main Success Scenario	1. Donor chooses 'Post Donation' option. 2. System renders donor to post donation form. 3. Donor enters all necessary information required to post donation. 4. System verifies all information entered by user. 5. Donor chooses submit option. 6. A new donation with available status is created and stored in database. 7. Donor receives confirmation message.
Extensions/Alternative Flows	**3A.** Incorrect information 1. System displays messages of incomplete information. 2. Donor enters correct information. **4A.** Incomplete information 1. System displays messages of incomplete information. 2. Donor enters complete information. **4B**. System failure 1. System gives the message of system failure. 2. Donation is not posted.
Frequency of Occurrence	medium
Inputs	1. Food type 2. Quantity 3. Expiry Date 4. Pickup Time 5. Pickup location
Outputs	1. Posted donation

Table 2.5: Post Donation

UC6: Edit Available Information

Table 2.6 shows the usecase description of Edit Available Donation, donor chooses option edit to edit the donations posted by donor and are still with available status and are not requested or booked or collected to the system, and the system displays the details of donation, then user submits the new information and system displays confirmation message and if information provided by donor is incomplete, then the system responds with the message of invalid detail.

ID	6
Name	Edit Available Information

Primary Actor	Donor
Stakeholders and Interests	**Donor:** wants to edit donations so that donation has no incorrect information. **Admin:** Wants to get accurate information of donor donation to handle platform. **NGO:** wants updated and correct donations to receive them.
Preconditions	1. Donor has logged in. 2. Donation already exists. 3. Donation status is available.
Postconditions	1. Donation is updated. 2. NGOs receive donation notification. 3. Ngo and admin can view.
Main Success Scenario	1. Donor chooses 'Edit Donation' option. 2. System renders donor to edit donation form. 3. Donor enters all necessary information required to update donation. 4. System verifies all information entered by user. 5. Donor chooses submit option. 6. Donation with available status is updated with new entered information and stored in database. 7. Donor receives confirmation message.
Extensions/Alternative Flows	**3A.** Incorrect information 1. System displays messages of incomplete information. 2. Donor enters correct information. **4A.** Incomplete information 1. System displays messages of incomplete information. 2. Donor enters complete information. **4B**. System failure 1. System gives the message of system failure. 2. Donation is not updated. **4C**. Donor edits non editable field 1. System shows the message field cannot be edited.
Frequency of Occurrence	medium
Inputs	1. Updated donation fields
Outputs	1. Updated donation

Table 2.6: Edit Available Donation

UC7: Delete Available Donation

Table 2.7 shows the usecase description of Delete Available Donation, donor chooses Delete Available Donation option to request to delete the donations posted by donor and are still with available status and are not requested or booked or collected to the system, and the system displays the message of confirmation and if donation provided by donor is other than with status of available, then the system displays the message of error.

ID	7

Name	Delete Available Donation
Primary Actor	Donor
Stakeholders and Interests	**Donor:** wants to delete donation that donor no longer intends to donate. **Admin:** Wants to get accurate information deleted donations. **NGO:** do not want to see donations that are no longer available.
Preconditions	1. Donor has logged in. 2. Donation already exists. 3. Donation status is available.
Postconditions	1. Donation is deleted. 2. NGO can no longer view it.
Main Success Scenario	1. Donor views the donation list. 2. Donor chooses the delete option on donation to delete in donation list. 3. System checks whether the donation is available or not. 4. System removes donations from systems and databases. 5. Donor receives confirmation message. 6. System updates the new donation list.
Extensions/Alternative Flows	**3A.** Donation with other than available status 1. System gives the message that cannot delete donation with available status. 2. Donation is not deleted. **4A**. System failure 1. System gives the message of system failure. 2. Donation is not deleted.
Frequency of Occurrence	low
Inputs	1. Donation id
Outputs	1. Donation is removed from the system

Table 2.7: Delete Available Donation

UC8: View Donor Donation History

Table 2.8 shows the usecase description of View Donor Donation history, donor chooses option Donation history to view history of all the donations posted by donor to the system, and the system displays the list of donations and then donor applies filter and the system displays the updated list of donations and if no donation to show is available, then the system responds with the message of error.

ID	8
Name	View Donor Donation History
Primary Actor	Donor
Stakeholders and Interests	**Donor:** wants to see all the donations posted by donor.
Preconditions	1. Donor has logged in. 2. At least one donation is posted by donor.

Postconditions	1. Donation list is viewable to donor.
Main Success Scenario	1. Donor chooses 'Donation history' option. 2. System retrieves all the donations posted by donor. 3. System organizes the donations by status (requested, booked, collected). 4. System shows all the donations. 5. Donor can apply filters of • Status • Expiry date 6. System sorts the donations according to the filter 7. Donor can view the donation list and its details
Extensions/Alternative Flows	**2A.** No donation history 1. System sends the message with no donation posted. **2B**. System failure 1. System gives the message of system failure. 2. Donation list is not displayed.
Frequency of Occurrence	high
Inputs	1. Donor id
Outputs	1. Donations list

Table 2.8: View Donor Donation History

UC-9: View Donor Profile

Table 2.9 shows the usecase description of view donor profile. The donor chooses my profile option to view donor profile, and the system displays the details of donor.

ID	9
Name	View Donor Profile
Primary Actor	Donor
Stakeholders and Interests	**Donor:** wants to see the profile details.
Preconditions	1. Donor has logged in.
Postconditions	1. Donor can view its profile.
Main Success Scenario	1. Donor chooses 'My Profile' option. 2. System retrieves all the information about that donor. 3. System organizes its information • User type • Name • Email • Password • Location

	• Contact number 4. System displays all the details of the donor. 5. Donor can view its profile.
Extensions/Alternative Flows	**2A**. System failure 1. System gives the message of system failure. 2. Donor profile is not displayed.
Frequency of Occurrence	medium
Inputs	1. Donor id
Outputs	1. Donor profile

Table 2.9: View Donor Profile

UC-10: Edit Donor Profile

Table 2.10 shows the usecase description of Edit Donor Profile, donor chooses Edit Profile option to update the profile of the donor, and the system displays the details of donor, then donor submits the new information and system displays confirmation message and if information provided by donor is incomplete, then the system displays the message of incomplete detail and if information provided by donor is invalid, then the system displays the message of invalid detail and if information provided by donor is not editable, then the system displays the message of it.

ID	10
Name	Edit Donor Profile
Primary Actor	Donor
Stakeholders and Interests	Donor: wants to edit its profile information
Preconditions	1. Donor has logged in.
Postconditions	1. Updated Profile information of donor.
Main Success Scenario	1. Donor chooses ‘Edit Profile’ option from view profile. 2. System displays form to edit the profile. 3. Donor enters all the information that needs to be edited. • Name • Email • Password • Location • Contact number 4. Donor chooses submit option. 5. System checks the information entered by the donor. 6. System updates the information in database.
Extensions/Alternative Flows	**3A.** Incorrect information 1. System displays messages of incomplete information. 2. Donor enters correct information. **4A.** Incomplete information

	1. System displays message of incomplete information. 2. Donor enters complete information. **4B**. System failure 1. System gives the message of system failure. 2. Profile is not updated. **4C**. Donor edits non editable field(email) 1. System shows the message field cannot be edited.
Frequency of Occurrence	low
Inputs	1. Profile information to update
Outputs	1. Updated profile information

Table 2.10: Edit Donor Profile

UC-11: NGO Signup

Table 2.11 shows the usecase description of NGO signup, NGO chooses signup option with details that are name, email, password, contact number and location and document, then the system creates new user with unverified account and if the signup information provided by user is incomplete , then the system displays the message of incomplete information and if the signup email provided by user is already exists, then the system displays the message of it.

ID	11
Name	NGO Signup
Primary Actor	NGO
Stakeholders and Interests	**NGO:** wants to quickly sign up to accept donations. **Admin:** Wants to get accurate information of NGOS to handle platform.
Preconditions	1. User has not already signed up. 2. System is working and connected to database.
Postconditions	1. A new NGO created with unverified status. 2. Documents sent to admin for verification.
Main Success Scenario	1. User selects the signup option. 2. User is redirected to the signup form. 3. User fills in all the information and the documents required in signup form. 4. User selects the submit option. 5. System checks the validation of information entered by users. 6. System creates new user with unverified status. 7. System gives messages of new user created with unverified status.
Extensions/Alternative Flows	**4A.** Incomplete information 1. System shows the message of incomplete information. 2. User enters the incomplete information.

	5A. Duplicate email. 1. System shows the message that email already exists. 2. User enters the new email. **5B**. Weak and Lower length password 1. System shows the message of weak and less length password. 2. User enters valid password. **6A**. System failure 1. System gives the message of system failure. 2. Account is not created.
Frequency of Occurrence	High
Inputs	1. User type 2. Organization Name 3. Email 4. Password 5. Location 6. Contact number 7. NGO registration document
Outputs	1. User created with unverified status 2. User id generated 3. User stored in database 4. Confirmation notification

Table 2.11: NGO Signup

UC-12: View Available Donations

Table 2.12 shows the usecase description of View Available Donation, NGO chooses view donations option to view all the donations that are posted by all donors and are available and can be requested to the system, and the system then displays the list of donations and if donation with available donations does not exist, then the system displays the message of error.

ID	12
Name	View Available donations
Primary Actor	NGO
Stakeholders and Interests	**NGO:** want to accept donations by viewing them. **Donor:** wants the donations to be accepted. **Admin:** wants to ensure fair access.
Preconditions	1. NGO is logged in.
Postconditions	1. NGO can view the list of available donations sorted by time. 2. NGO can request those donations.
Main Success Scenario	1. NGO selects the 'Available Donations' option. 2. System retrieves all the donations posted by all donors with available status. 3. System organizes the donations sorted chronologically.

	4. System displays each donation with information • Food type • Quantity • Expiry Date • Pickup Time • Pickup location 5. NGO can view all the donations with available status.
Extensions/Alternative Flows	**2A.** No donations available 1. System displays the message of no available donations. **6A**. System failure 1. System gives the message of system failure. 2. No donations are displayed.
Frequency of Occurrence	High
Inputs	1. NGO authentication token
Outputs	1. List of available donations sorted by time.

Table 2.12: View Available Donations

UC-13: View request History

Table 2.13 shows the usecase description of View request History, NGO chooses request history option to view the donations requested to book so NGO can collect them to the system, and the system responds with the list of donations and if donation with available donations does not exist, then the system displays the message of error.

ID	13
Name	View Request History (Pending, Approved and Reject)
Primary Actor	NGO
Stakeholders and Interests	**NGO**: wants to view all the donations NGO requested for and were they accepted, rejected or still pending.
Preconditions	1. NGO has logged in. 2. NGO has made at least one request for the donation.
Postconditions	1. NGO can view the list of all the donations it requested for and the request status.
Main Success Scenario	1. NGO selects 'Request History' option. 2. System retrieves all the donations for which NGO has requested. 3. System organizes the donations sorted by the status of request (pending, approved or rejected). 4. System displays each donation with information • Food type • Quantity • Expiry Date • Pickup Time • Pickup location • Request status

	• Donation status 5. NGO can view all the donations with available status.
Extensions/Alternative Flows	**2A.** No requested donations are available 1. System displays the message of no available requested donations. **6A**. System failure 1. System gives the message of system failure. 2. No donations are displayed.
Frequency of Occurrence	High
Inputs	1. NGO id
Outputs	1. List of NGOS requested donations

Table 2.13: View Request History

UC-14: Request Available Donation

Table 2.14 shows the usecase description of Request Available Donation, NGO chooses request option to request donation that are currently available to the system, and the system creates the request by setting the status of donation as requested and if request was already made by the NGO, then the system displays the message of it.

ID	14
Name	Request Available Donation
Primary Actor	NGO
Stakeholders and Interests	**NGO:** want to request donations to collect them. **Donor:** wants the donations to be accepted. **Admin:** wants to ensure fair access.
Preconditions	1. NGO has logged in. 2. Donations exist with available status. 3. NGO already has not requested it. 4. NGO has available slot to request donation.
Postconditions	1. Requests are created with pending status. 2. Donation status is requested. 3. Donation is sent to admin for approval. 4. Approval notification is sent to admin
Main Success Scenario	1. NGO selects ‘request’ option on available donation. 2. System validates all pre-conditions. 3. System creates request with status pending. 4. System updates status to requested. 5. System adds request to NGO request history.
Extensions/Alternative Flows	**2A.** Donation is already requested 1. System displays message request already maded **3A**. System failure 1. System gives the message of system failure. 2. Donation is not requested.
Frequency of Occurrence	High
Inputs	1. NGO id

	2. Donation id
Outputs	1. Request status with pending 2. Donation status is requested 3. Approval notification for admin.

Table 2.14: Request Available Donation

UC-15: Mark Booked Donations as Collected

Table 2.15 shows the usecase description of Mark Booked Donations as Collected, NGO views all the donations that NGO booked to collect and then for a booked donation chooses collect option to mark it collected after collecting it from pickup location, and the system responds by marking that donation as collected

ID	15
Name	Mark Booked Donations as Collected
Primary Actor	NGO
Stakeholders and Interests	**NGO:** wants delivered and booked donation to be marked collected. **Donor:** wants delivered and booked donation to be marked collected. **Admin:** wants transparent tracking of the status.
Preconditions	1. NGO has logged in. 2. Donations exist with booked status. 3. NGO has physically collected the donation.
Postconditions	1. Donation status is changed to collected. 2. Donor is notified of collection.
Main Success Scenario	1. NGO views the booked donation list. 2. System shows Collected option on donations. 3. NGO selects collected option on the donation to mark it collect. 4. System updates the donation status to collected. 5. System send notification donor for conformation.
Extensions/Alternative Flows	**3A**. System failure 1. System gives the message of system failure. 2. Donation is not marked collected
Frequency of Occurrence	Medium
Inputs	1. Donation id 2. NGO id
Outputs	1. Donation status is changed to collected.

Table 2.15: Mark Booked Donations as Collected

UC-16: View NGO Donation history

Table 2.16 shows the usecase description of View NGO Donation history, NGO chooses option Donation history to view history of all the donations posted by NGO to the system, and the system displays the list of donations and then NGO applies filter and the system displays the updated list of donations and if no donation to show is available, then the system displays the message of error.

ID	16
Name	View NGO Donation history
Primary Actor	NGO
Stakeholders and Interests	**NGO**: want to see all donation activities.
Preconditions	1. NGO has logged in. 2. NGO has made at least request for the donation.
Postconditions	1. NGO views all donations with their information and statuses.
Main Success Scenario	1. NGO selects 'Donation History' option. 2. System retrieves all the donations related to the NGO 3. System organizes the donations by status (requested, booked, collected). 4. System shows all the donations. 5. NGO can apply filters of • Status • Expiry date 6. System sorts the donations according to the filter 7. NGO can view the donation list and its details
Extensions/Alternative Flows	**2A.** No donation history 2. System sends the message with no donation posted. **2B**. System failure 3. System gives the message of system failure. 4. Donation list is not displayed.
Frequency of Occurrence	High
Inputs	1. NGO id
Outputs	1. List of donations

Table 2.16: View NGO Donation History

UC-17: View NGO Profile

Table 2.17 shows the usecase description of view NGO profile. The NGO chooses my profile option to view NGO profile, and the system displays the details of NGO.

ID	17
Name	View NGO Profile
Primary Actor	NGO

Stakeholders and Interests	**NGO:** wants to see the profile details.
Preconditions	1. NGO has logged in.
Postconditions	1. NGO can view its profile.
Main Success Scenario	1. NGO chooses 'My Profile' option. 2. System retrieves all the information about that NGO. 3. System organizes its information • User type • Name • Email • Password • Location • Contact number • Documents 4. System displays all the details of the NGO. 5. NGO can view its profile.
Extensions/Alternative Flows	**2A**. System failure 1. System gives the message of system failure. 2. NGO profile is not displayed.
Frequency of Occurrence	medium
Inputs	1. NGO id
Outputs	1. NGO profile

Table 2.17: View NGO Profile

UC-18: Edit Donor Profile

Table 2.18 shows the usecase description of Edit NGO Profile, NGO chooses Edit Profile option to update the profile of the NGO, and the system displays the details of NGO, then NGO submits the new information and system displays confirmation message and if information provided by NGO is incomplete, then the system displays the message of incomplete detail and if information provided by NGO is invalid, then the system displays the message of invalid detail and if information provided by donor is not editable, then the system displays the message of it.

ID	18
Name	Edit NGO Profile
Primary Actor	NGO
Stakeholders and Interests	**NGO**: wants to edit its profile information
Preconditions	1. NGO has logged in.
Postconditions	1. Updated Profile information of NGO.
Main Success Scenario	1. NGO chooses 'Edit Profile' option from view profile.

	2. System displays form to edit the profile. 3. NGO enters all the information that needs to be edited. • Name • Email • Password • Location • Contact number 4. NGO chooses submit option. 5. System checks the information entered by the NGO. 6. System updates the information in database. 7. System sends confirmation messages to user.
Extensions/Alternative Flows	**3A.** Incorrect information 1. System displays messages of incomplete information. 2. NGO enters correct information. **4A.** Incomplete information 1. System displays messages of incomplete information. 2. NGO enters complete information. **4B**. System failure 1. System gives the message of system failure. 2. Profile is not updated. **4C**. NGO edits non editable field (documents, email) 1. System shows the message field cannot be edited.
Frequency of Occurrence	low
Inputs	2. Profile information to update
Outputs	2. Updated profile information

Table 2.18: Edit NGO Profile

UC-19: Verify NGOs Signup

Table 2.19 shows the usecase description of Verify NGOs Signup, Admin chooses pending verifications to view all the pending verification requests of NGOs of Signup so that there account is verified to login, and on this the system displays the list of NGOs, then Admin chooses view NGO option to view the details of NGO and system shows the detail of NGOs and then NGO approves or rejects the NGO and system sets the NGOs status approved or rejected.

ID	19
Name	Verify NGOs Signup
Primary Actor	Admin
Stakeholders and Interests	**Admin:** want to verify NGOs Signup. **NGO:** wants quick verification to receive donations. **Donor:** want authenticate NGOs. **System:** wants verified NGOs to make system secure
Preconditions	1. Admin has logged in.

	2. At least NGO has made signup request. 3. NGO has uploaded documents.
Postconditions	1. NGO account status is updated to verified. 2. NGO can log in.
Main Success Scenario	1. Admin logs into admin dashboard. 2. System displays 'Pending verifications'. 3. Admin chooses 'pending verification' option. 4. System displays a list of NGOs with unverified status and review option. 5. Admin choose review option. 6. System opens NGO detail unchangeable information • Organization name • Organization email • Contact number • Uploaded document • Approve option • Reject option 7. Admin reviews the documents and other information. 8. Admin chooses approve option. 9. System updates NGO status to verified. 10. System sends notification to NGO
Extensions/Alternative Flows	3A. No NGOs to verify 1. System displays the message that there are no NGOs to verify. 8A. Illegible NGO 1. Admin chooses reject button. 2. System shows the list of reasons to reject • Incomplete documents • Illegible documents • Invalid document 3. Amin chooses one option 4. Admin selects ok option 5. NGO status is changed to rejected 6. Notification is sent to NGO
Frequency of Occurrence	Medium
Inputs	1. NGO Signup request id 2. NGO uploaded documents 3. Verification decision(approve /reject) 4. Comment 5. Admin authentication token
Outputs	1. NGO account status updated (verified/rejected) 2. Confirmation notification to NGO

Table 2.19: Verify NGO Signups

UC-20: Delete User

Table 2.20 shows the usecase description of Delete User, Admin choses user management to view all the users of the system donor and NGOs, and the system shows the list of Users and then Admin sends chooses delete option to delete a user and the system shows confirmation message after deleting it.

ID	20
Name	Delete User
Primary Actor	Admin
Stakeholders and Interests	**Admin**: wants to maintain platform integrity by keep only active and responsible users.
Preconditions	1. Admin has logged in. 2. Users exist in the system.
Postconditions	1. User account is removed from database.
Main Success Scenario	1. User selects 'User management' option. 2. System retrieves all the users. 3. System organizes them according to user type. • NGO • Donor 4. System displays the list of users with information • User type • Name • Email • Contact • Delete option 5. User selects delete option. 6. System removes the user from database. 7. System updates all reference to maintain data
Extensions/Alternative Flows	2A. No Users in database 1. System displays the message of no user
Frequency of Occurrence	low
Inputs	1. User id
Outputs	1. User is removed from database

Table 2.20: Delete User

UC-21: View Users List

Table 2.21 shows the usecase description of View Users List, Admin choses user management to view all the users of the system donor and NGOs, and the system shows the list of Users organized by NGO and Donor and admin selects specific user and system displays their details.

ID	21

Name	View Users List
Primary Actor	Admin
Stakeholders and Interests	**Admin**: wants to view all users of the system.
Preconditions	3. Admin has logged in. 4. Users exist in the system.
Postconditions	2. Admin can view list of users.
Main Success Scenario	8. User selects 'User management' option. 9. System retrieves all the users. 10. System organizes them according to user type. • NGO • Donor 11. System displays the list of users with information • User type • Name • Email • Contact • View option 12. User selects View option. 13. System shows detailed information of user If Donor: • User type • Name • Email • Password • Location • Contact number If NGO: • User type • Organization Name • Email • Password • Location • Contact number • NGO registration document
Extensions/Alternative Flows	2A. No Users in database 2. System displays the message of no user
Frequency of Occurrence	low
Inputs	2. User id
Outputs	2. List of users

Table 2.21: View Users List

UC-22: View All Donations

Table 2.22 shows the usecase description of View All Donation, Admin chooses all donations option to view all the donation of the system, and the system displays the list of donors and if no donation to show is available, then the system message the message of error.

ID	22
Name	View all donations
Primary Actor	Admin
Stakeholders and Interests	**Admin**: wants to view all the donations to keep track o the system
Preconditions	1. Admin has logged in. 2. At least one donation is posted by any donor.
Postconditions	1. Admin views the list of all donations.
Main Success Scenario	1. Admin chooses 'All Donations' option. 2. System retrieves all the donations in database. 3. System organizes them according to donation status • Available • Requested • Booked • Collected 4. System shows the donation with information • Food type • Quantity • Expiry Date • Pickup Time • Pickup location 5. Admin view the donation
Extensions/Alternative Flows	2A. No donation available 1. System shows the message of no donations available
Frequency of Occurrence	Medium
Inputs	N/A
Outputs	1. Donation list

Table 2.22: View All Donations

UC-23: View NGO requests

Table 2.23 shows the usecase description of View NGOs Request, Admin chooses donation requests to view all request of NGOs for booking a donations, and the system displays the list of donation requests of NGOs and if no donation to show is available, then the system displays the message of error.

ID	23
Name	View NGOs requests
Primary Actor	Admin
Stakeholders and Interests	**Admin**: want to make donation decisions. **NGO**: want fair consideration for donation requests. **Donor**: want to donate as soon as possible.
Preconditions	1. Admin has logged in. 2. At least one NGO request exists.
Postconditions	1. Admin views all the NGOs request list.
Main Success Scenario	1. Admin chooses 'Donation Requests' option. 2. System retrieves all the donations with request status 3. System organizes them according to latest request time. 4. System displays them with information • Food type • Quantity • Expiry Date • Pickup Time • Pickup location 5. User can apply filters of • Food type • Quantity • Expiry Date • Pickup Time • Pickup location 6. Admin views NGOs requests
Extensions/Alternative Flows	3B. No donations are requested by NGOs 1. System displays message that there are no requests.
Frequency of Occurrence	High
Inputs	1. Donation id
Outputs	1. List of donations request

Table 2.23: View NGOs Requests

UC-24: Accept Requested Donation

Table 2.3 shows the usecase description of Accept Requested Donation, Admin chooses approve donation to accept request of NGOs for booking a donation, ,admin approves any one NGO for a donation and system approves the booking of that donation of NGO and auto rejecting all other NGOs requests and if Ngo does not have capacity to accept the donation , then the system responds with the message of error.

ID	24
Name	Accept requested donation
Primary Actor	Admin

Stakeholders and Interests	**Admin**: want to make decisions. **NGO**: want fair consideration for donation requests. **Donor**: want to donate as soon as possible.
Preconditions	1. Admin has logged in. 2. At least one NGO request exists.
Postconditions	1. One NGO is approved. 2. Donation status becomes booked. 3. All other NGOs are auto rejected.
Main Success Scenario	1. After viewing the request, there is option of 'Approve NGO' option on all donations. 2. Admin chooses the option for that donation he wants to choose. 3. System approves for that donation. 4. System sends notification to NGOs whose requests are accepted. 5. System auto rejects other NGOs requests. 6. Donation status is updated to booked.
Extensions/Alternative Flows	3A. Approved NGO does not has collection capacity 1. System gives message this NGO cannot accept more donations
Frequency of Occurrence	High
Inputs	NGO id
Outputs	1. Donation status becomes booked.

Table 2.24: Accept Requested Donation

UC-25: Manage Donation Status

Table 2.3 shows the usecase description of Manage Donation Status in which admin can change the status of donations to any other status ,Admin views a donation and then chooses status option and selects the updated status, and the updates the status and if the previous status of donation was available , then the system responds with the message of error.

ID	25
Name	Manage Donation Status
Primary Actor	Admin
Stakeholders and Interests	**Admin**: want to maintain system integrity. **NGO**: need accurate status information. **Donor**: need accurate status information.
Preconditions	1. Admin has logged in. 2. Donations exist in system. 3. Donation status is not available
Postconditions	1. Donation status is updated.
Main Success Scenario	1. Admin views donation detail. 2. Admin selects status option on donation.

	3. System displays list of donation options • Available • Requested • Booked • Collected 4. Admin chooses any option. 5. System updates the status of donation. 6. System make relative changes to that status.
Extensions/Alternative Flows	3A. Admin changes status from available to other. 1. System displays message of Invalid action.
Frequency of Occurrence	Low
Inputs	1. Donation id
Outputs	1. Status is updated

Table 2.25: Managing Donation Status

2.6 System Sequence Diagrams

UC-1: Login

Figure 2.3 shows the system sequence diagram of login for the user sending login request with inputs email and password to the system, and the user is then redirected to respective user module and if user adds invalid information it gives error message of invalid credentials.

Figure 2.1: Login

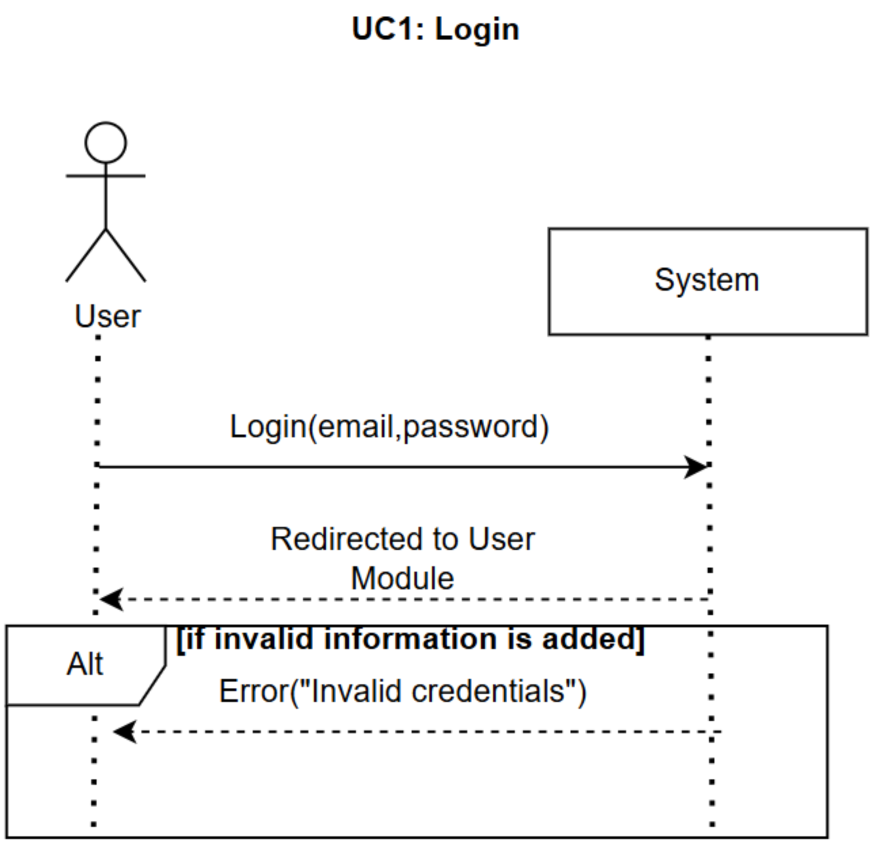


Figure 2.3: Login

UC-2: Logout

Figure 2.4 shows the system sequence diagram of logout for the user sending logout request to the system, and the user is then redirected to login page.

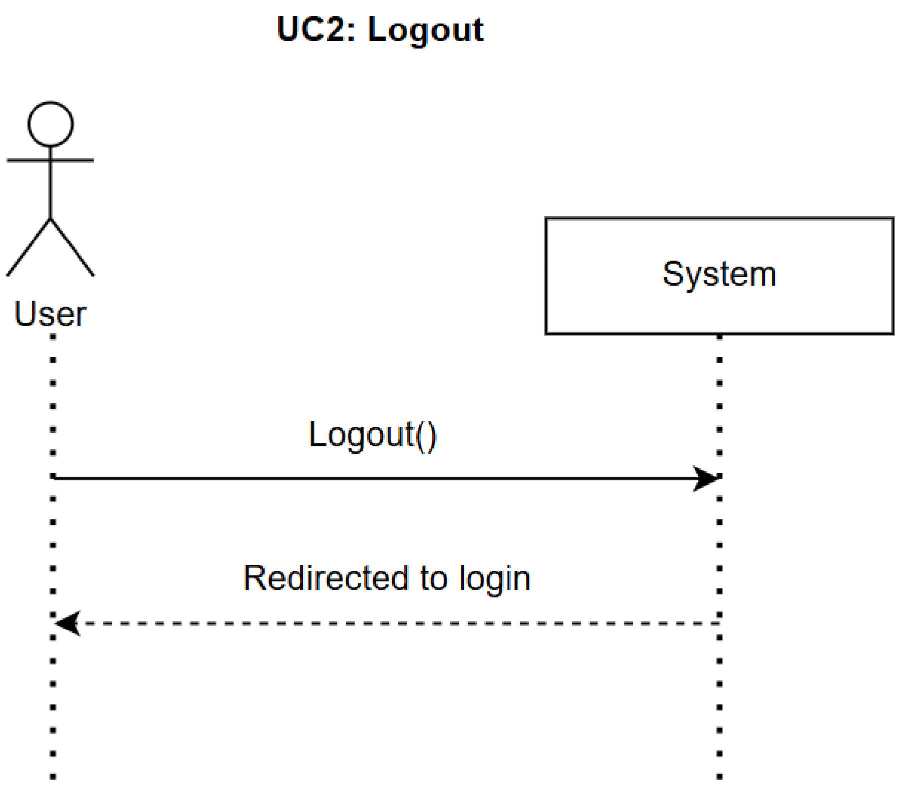


Figure 2.4: Logout

UC-3: View notifications

Figure 2.5 shows the system sequence diagram of view notifications, user sends view notifications request to the system, and the system returns the list of all the notifications and if no notification in the system, then the system responds with the message of no notifications to view.

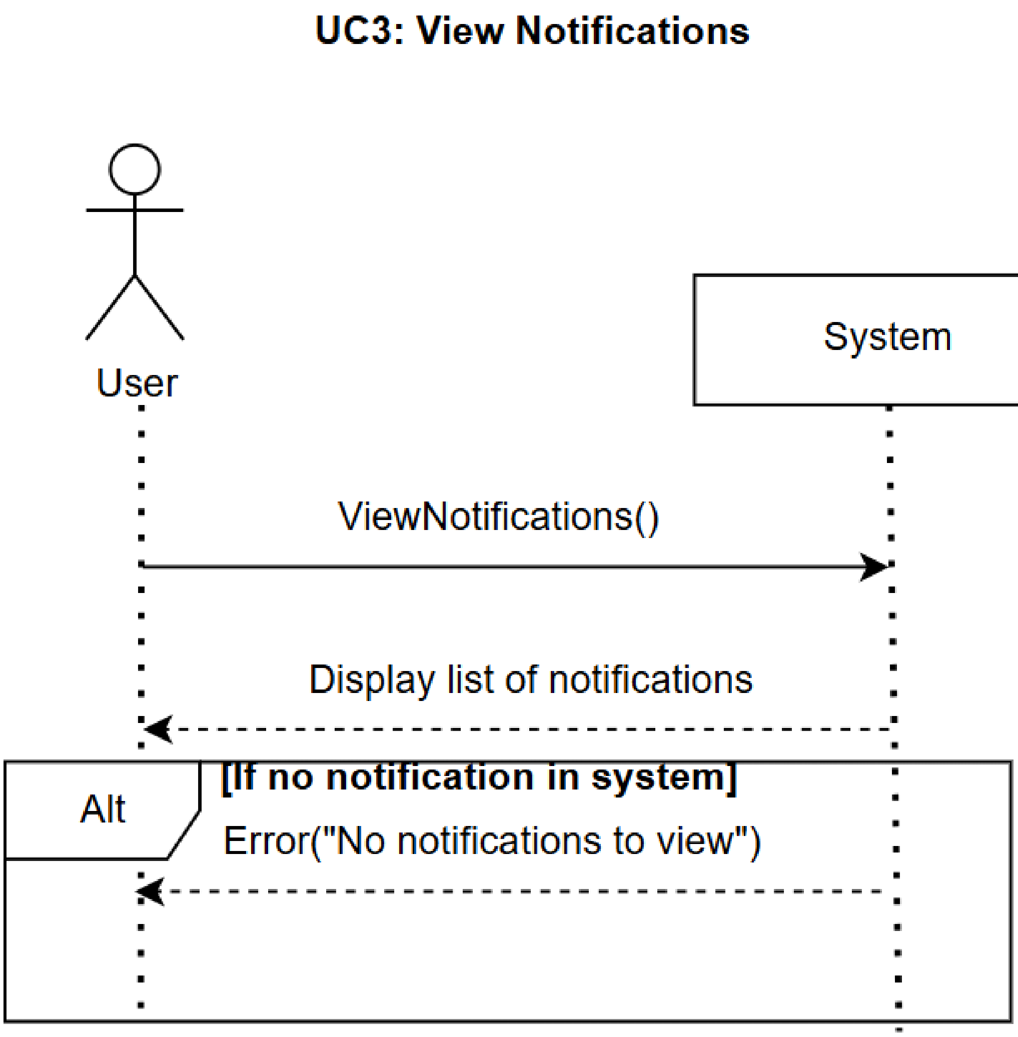


Figure 2.5: View Notification

UC-4: Donor Signup

Figure 2.6 shows the system sequence diagram of donor signup, donor sends the request of signing up with signup details that are name, email, password, contact number and location, then the system redirects the user to login page and if the signup information provided by user is incomplete , then the system responds with the message of incomplete information and if the signup email provided by user is already exists, then the system responds with the message of it.

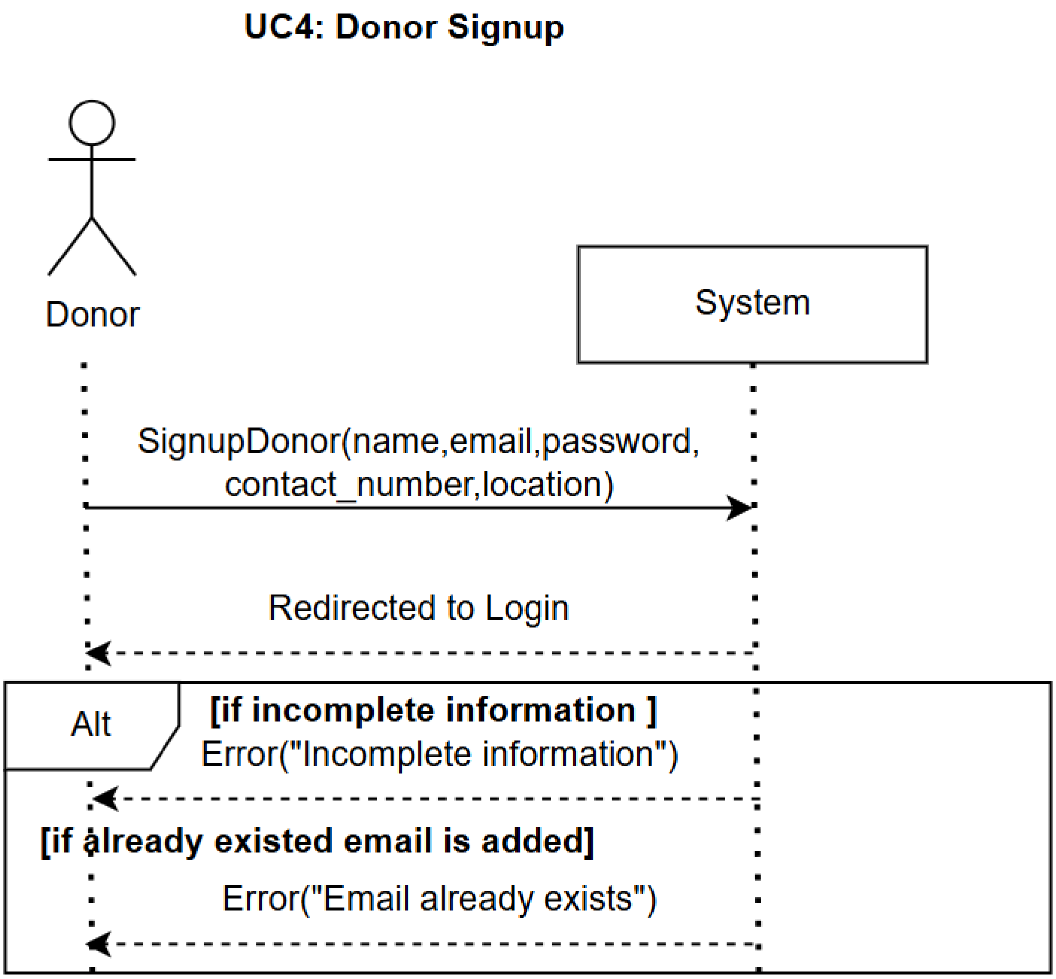


Figure 2.6: Donor Signup

UC-5: Post Donation

Figure 2.7 shows the system sequence diagram of post donation, donor sends post donation request to the system, and the system responds with the message of donation posted and if information provided by donor is incomplete, then the system responds with the message of invalid detail.

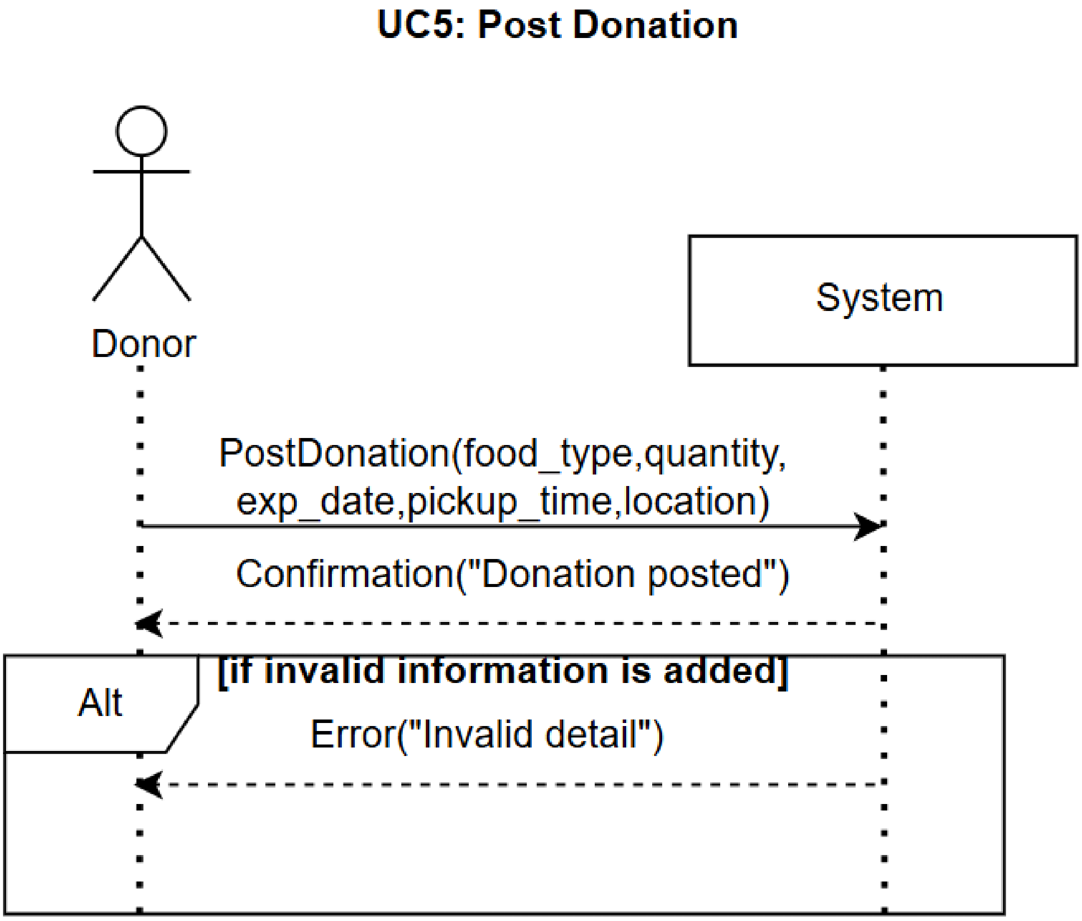


Figure 2.7: Post Donation

UC-6: Edit Available Donation

Figure 2.8 shows the system sequence diagram of Edit Available Donation, donor request to edit the donations posted by donor and are still with available status and are not requested or booked or collected to the system, and the system returns the details of donation, then user submits the new information and system responds with confirmation message and if information provided by donor is incomplete, then the system responds with the message of invalid detail.

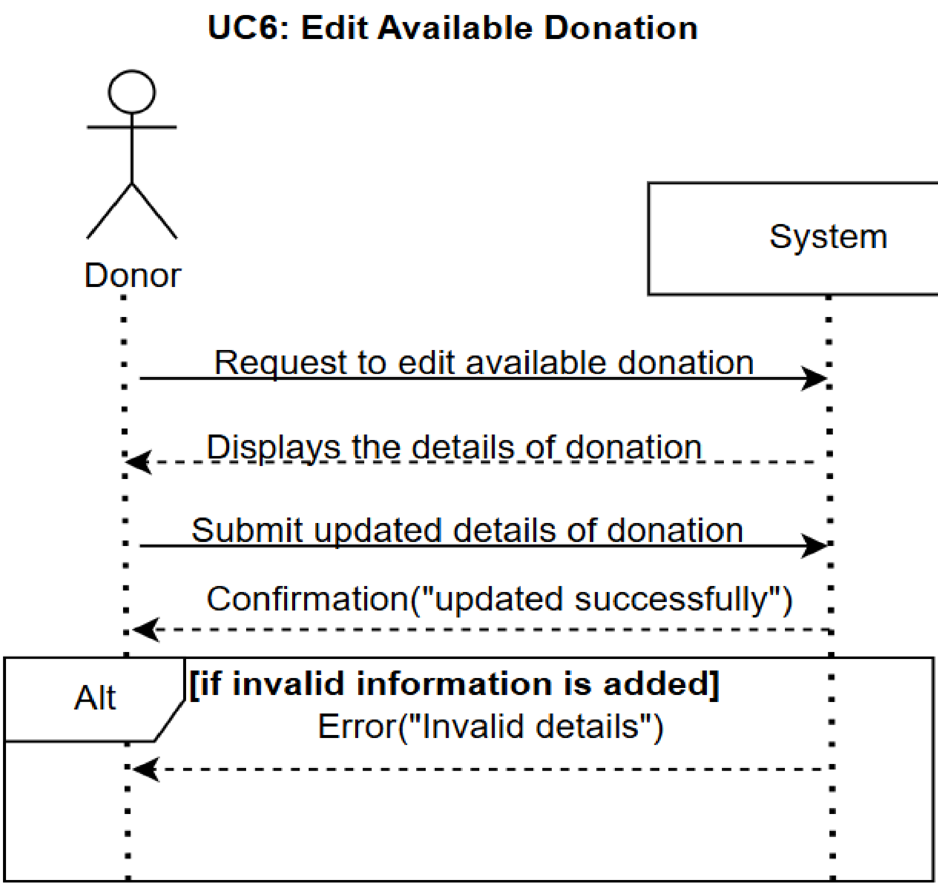


Figure 2.8: Edit Available Donation

UC-7: Delete Available Donation

Figure 2.9 shows the system sequence diagram of Delete Available Donation, donor sends request to delete the donations posted by donor and are still with available status and are not requested or booked or collected to the system, and the system responds with the message of confirmation and if donation provided by donor is other than with status of available, then the system responds with the message of error.

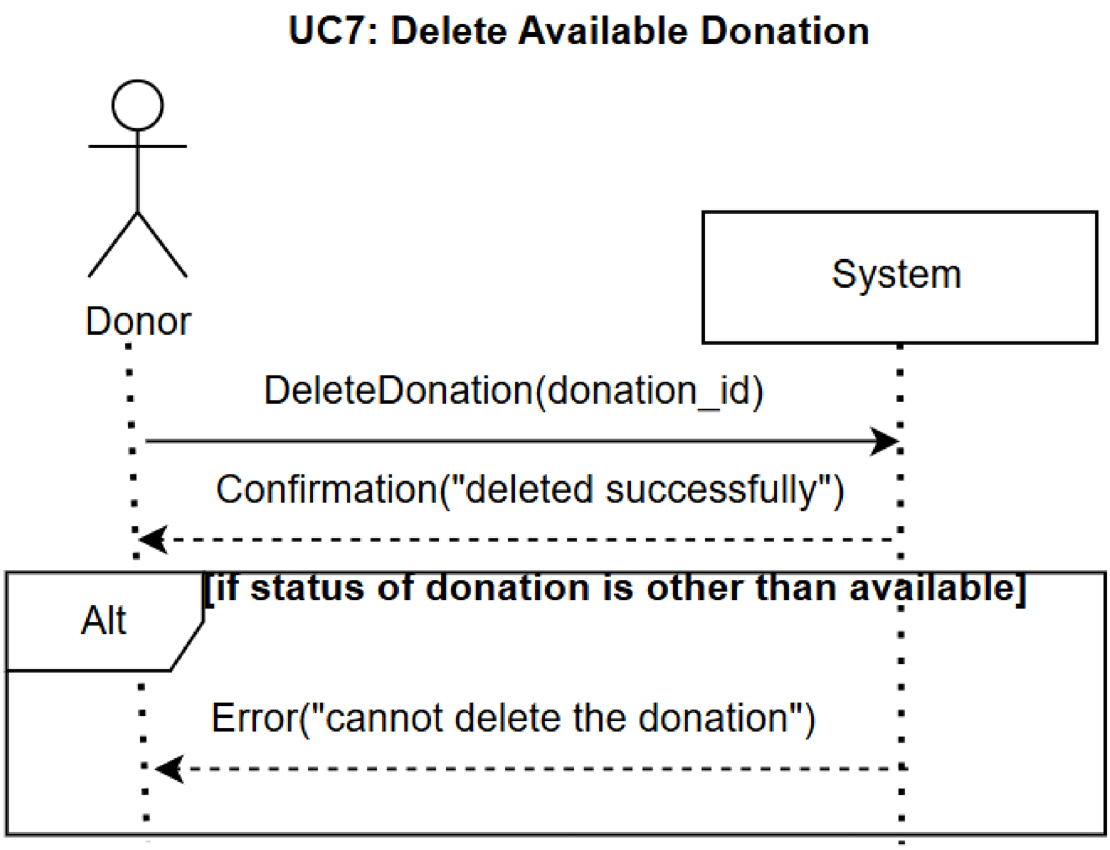


Figure 2.9: Delete Available Donation

UC-8: View Donor Donation history

Figure 2.10 shows the system sequence diagram of View Donor Donation history, donor make request view history of all the donations posted by donor to the system, and the system responds with the list of donations and then donor applies filter and the system responds with the updated list of donations and if no donation to show is available, then the system responds with the message of error.

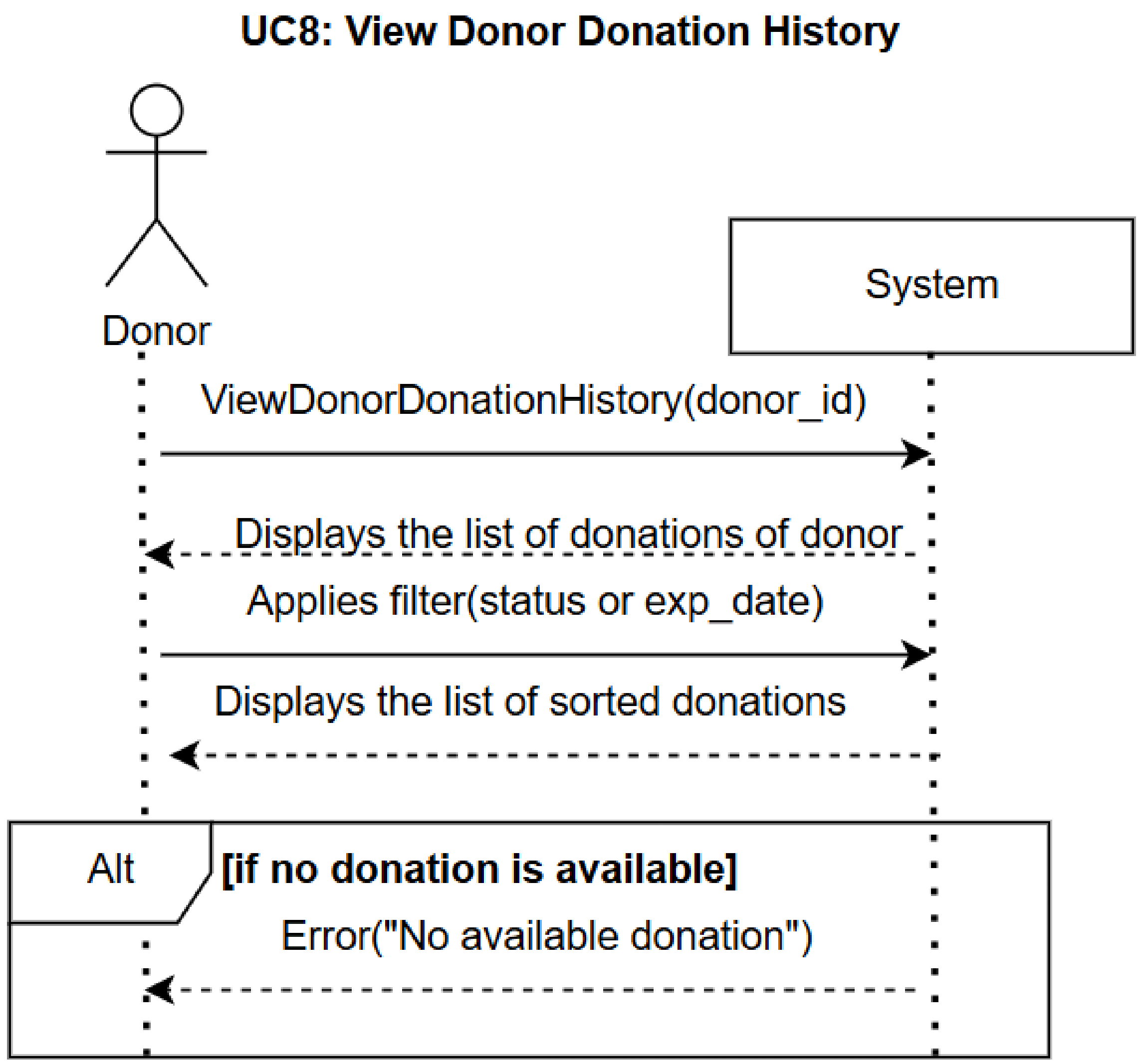


Figure 2.10: View Donor Donation History

UC-9: View Donor Profile

Figure 2.11 shows the system sequence diagram of view donor profile for the donor sending view donor profile request to the system, and the system responds with the details of donor.

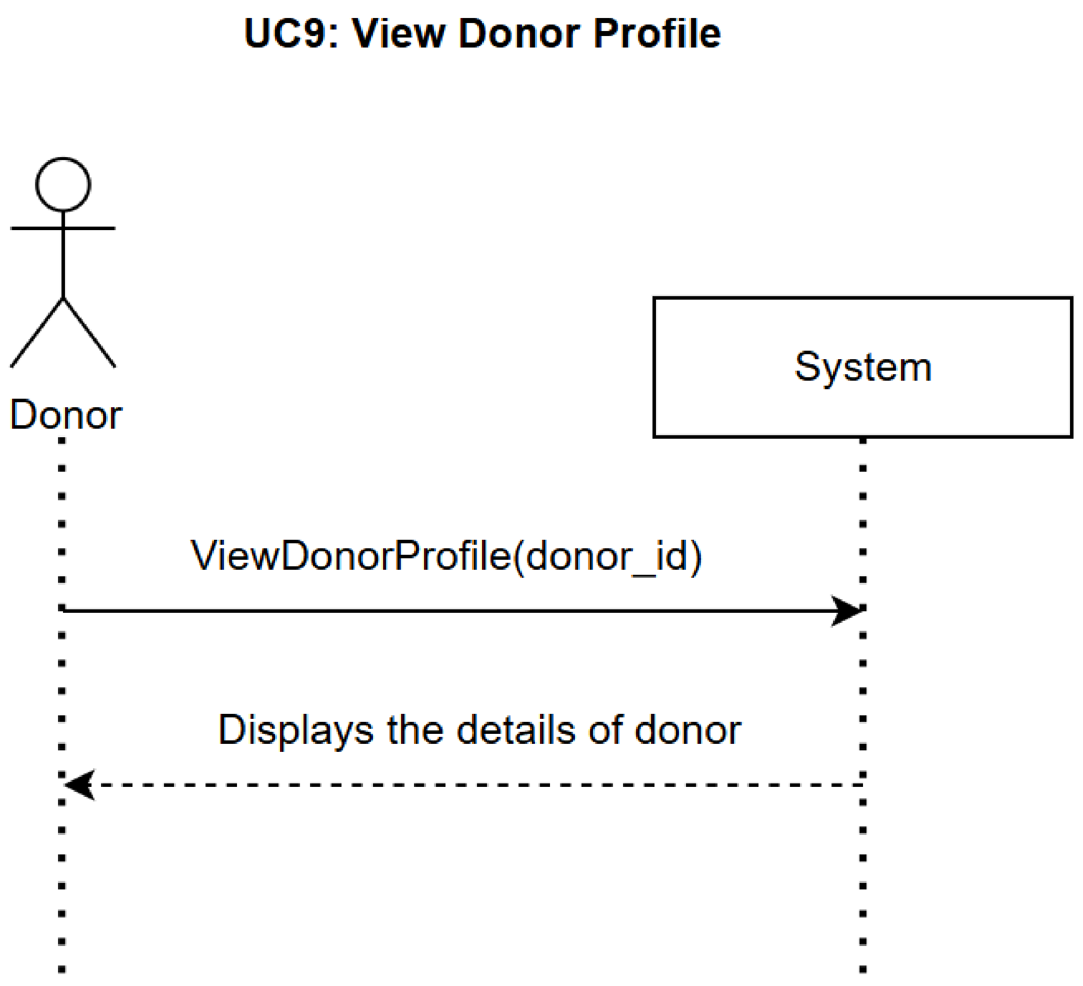


Figure 2.11: View Donor Profile

UC-10: Edit Donor Profile

Figure 2.12 shows the system sequence diagram of Edit Donor Profile, donor make Edit Donor Profile request to the system, and the system returns the details of donor, then donor submits the new information and system responds with confirmation message and if information provided by donor is incomplete, then the system responds with the message of incomplete detail and if information provided by donor is invalid, then the system responds with the message of invalid detail and if information provided by donor is not editable, then the system responds with the message of it.

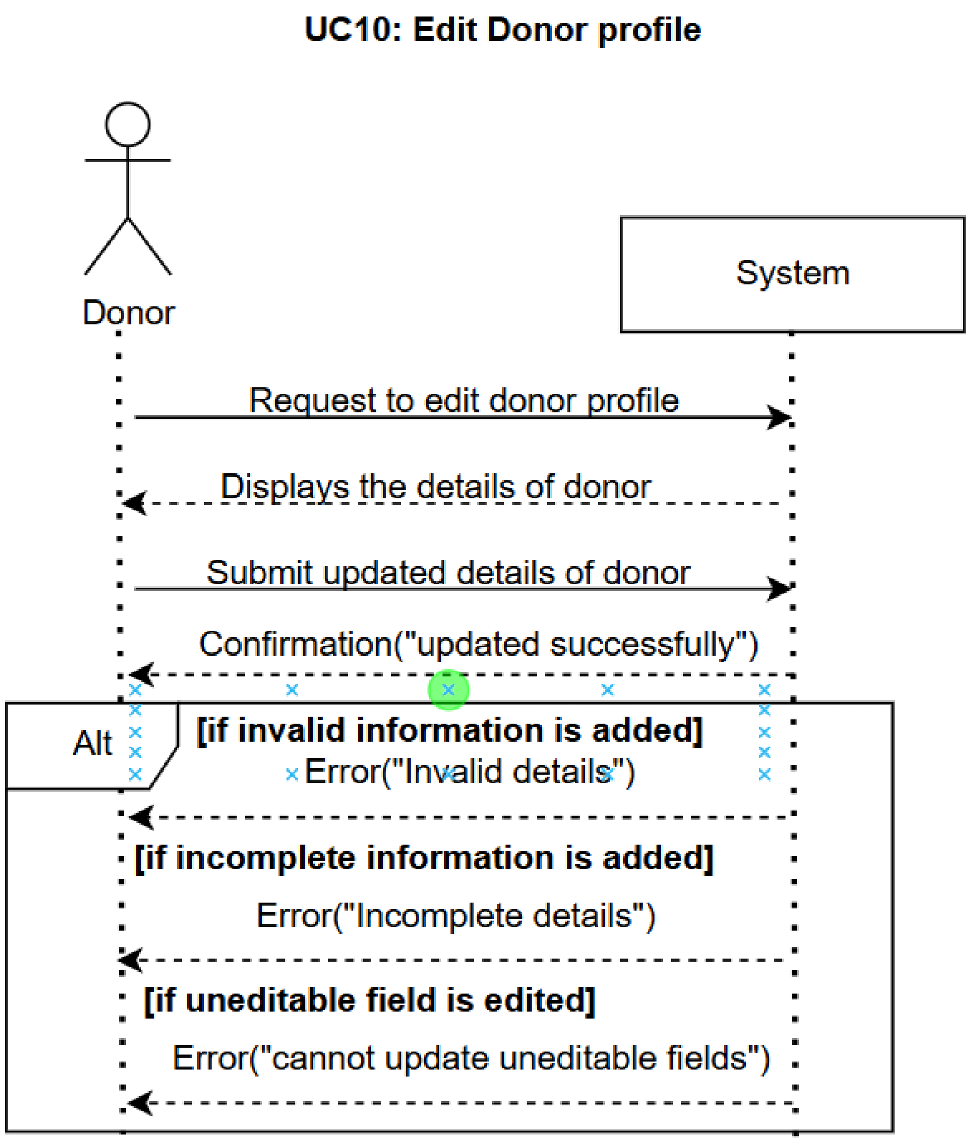


Figure 2.12: Edit Donor Profile

UC-11: NGO Signup

Figure 2.13 shows the system sequence diagram of NGO signup, NGO sends the request of signing up with signup details that are name, email, password, contact number and location and documents, then the system responds with confirmation message and if the signup information provided by NGO is incomplete , then the system responds with the message of incomplete information and if the signup email provided by NGO is already exists, then the system responds with the message of it.

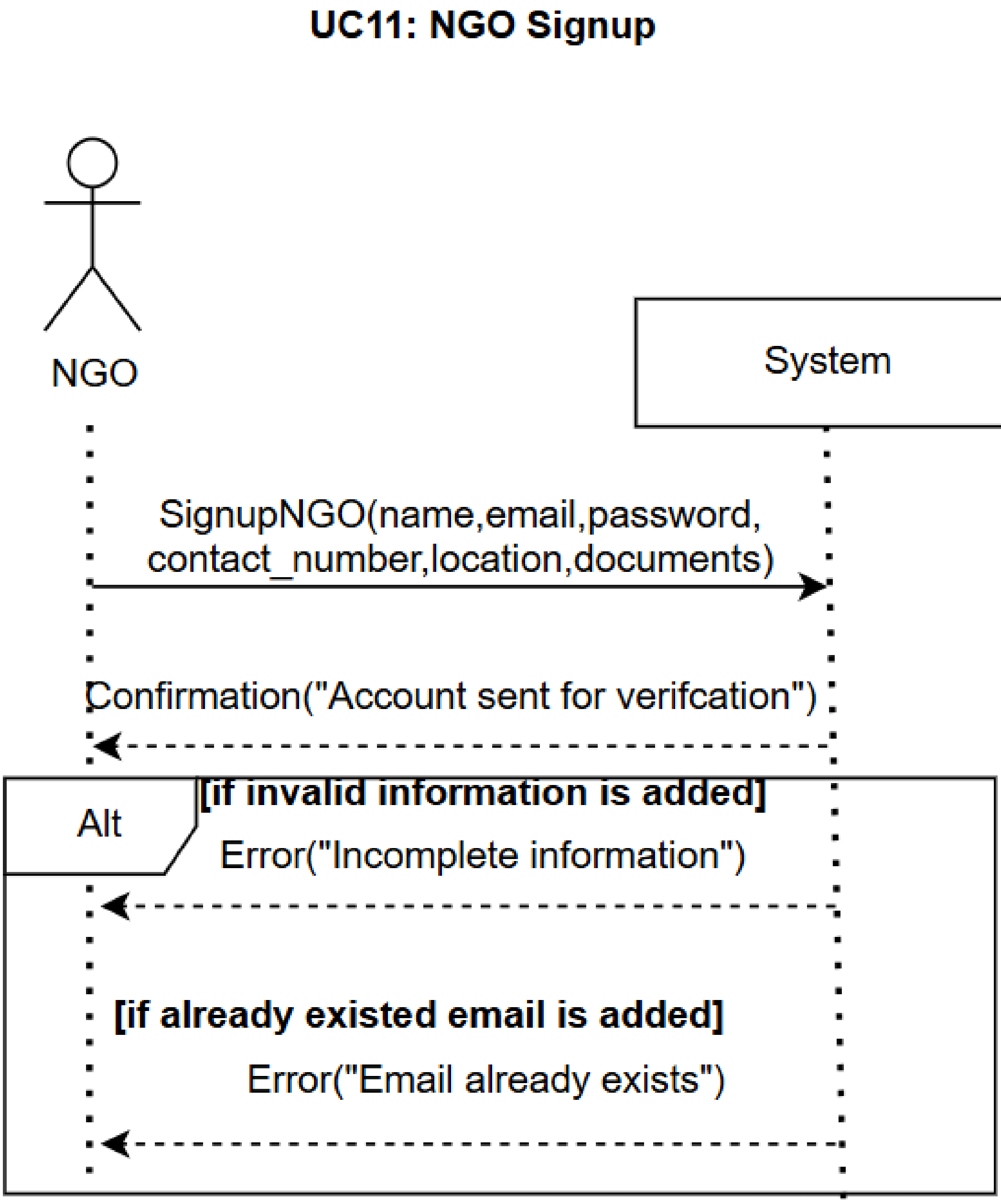


Figure 2.13: NGO Signup

UC-12: View Available Donations

Figure 2.14 shows the system sequence diagram of View Available Donation, NGO make request view all the donations that are posted by all donors and are available and can be requested to the system, and the system responds with the list of donations and if donation with available donations does not exist, then the system responds with the message of error.

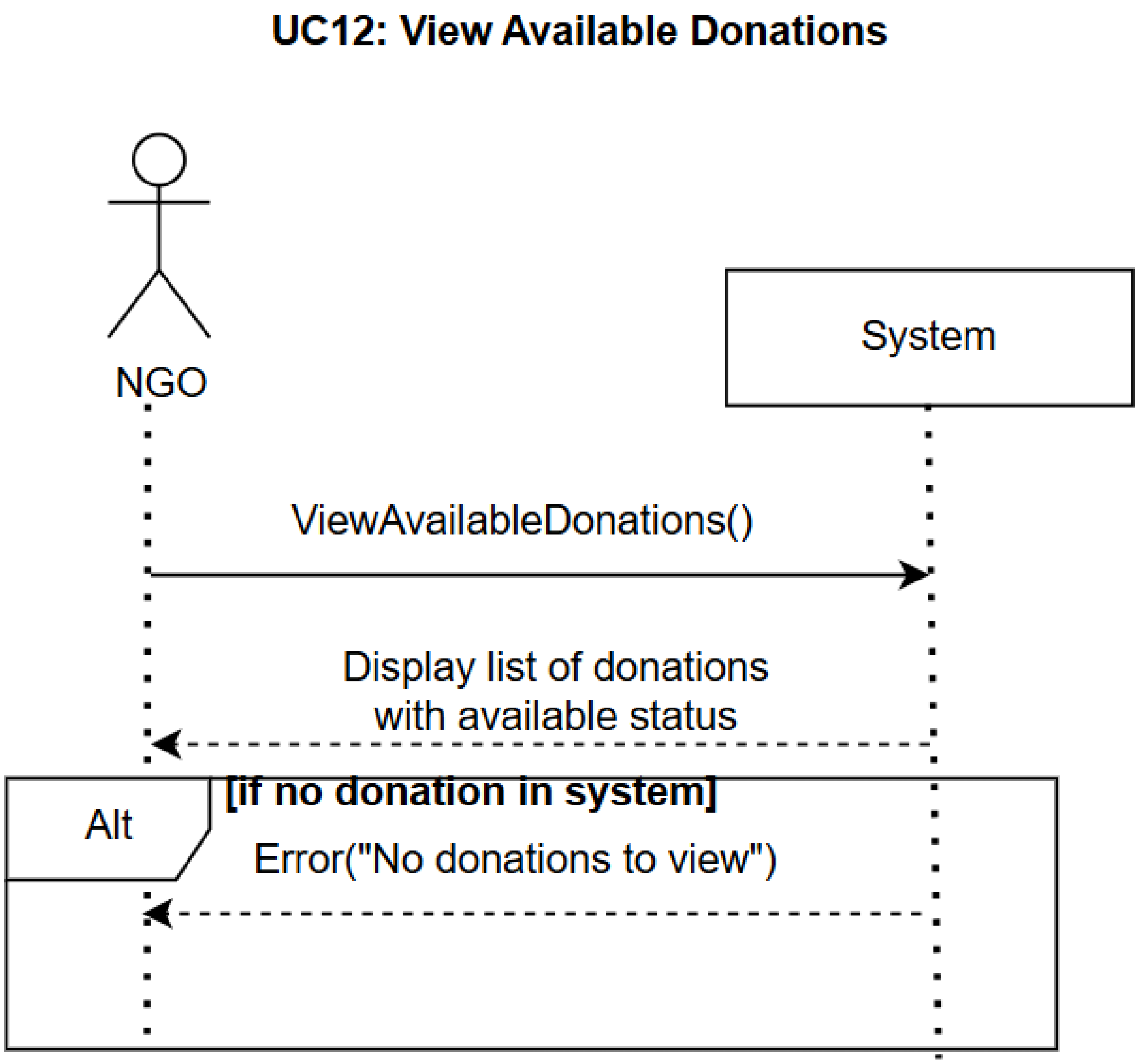


Figure 2.14: View Available Donations

UC-13: View request History

Figure 2.15 shows the system sequence diagram of View request History, NGO make request to view the donations requested to book so NGO can collect them to the system, and the system responds with the list of donations and if donation with available donations does not exist, then the system responds with the message of error.

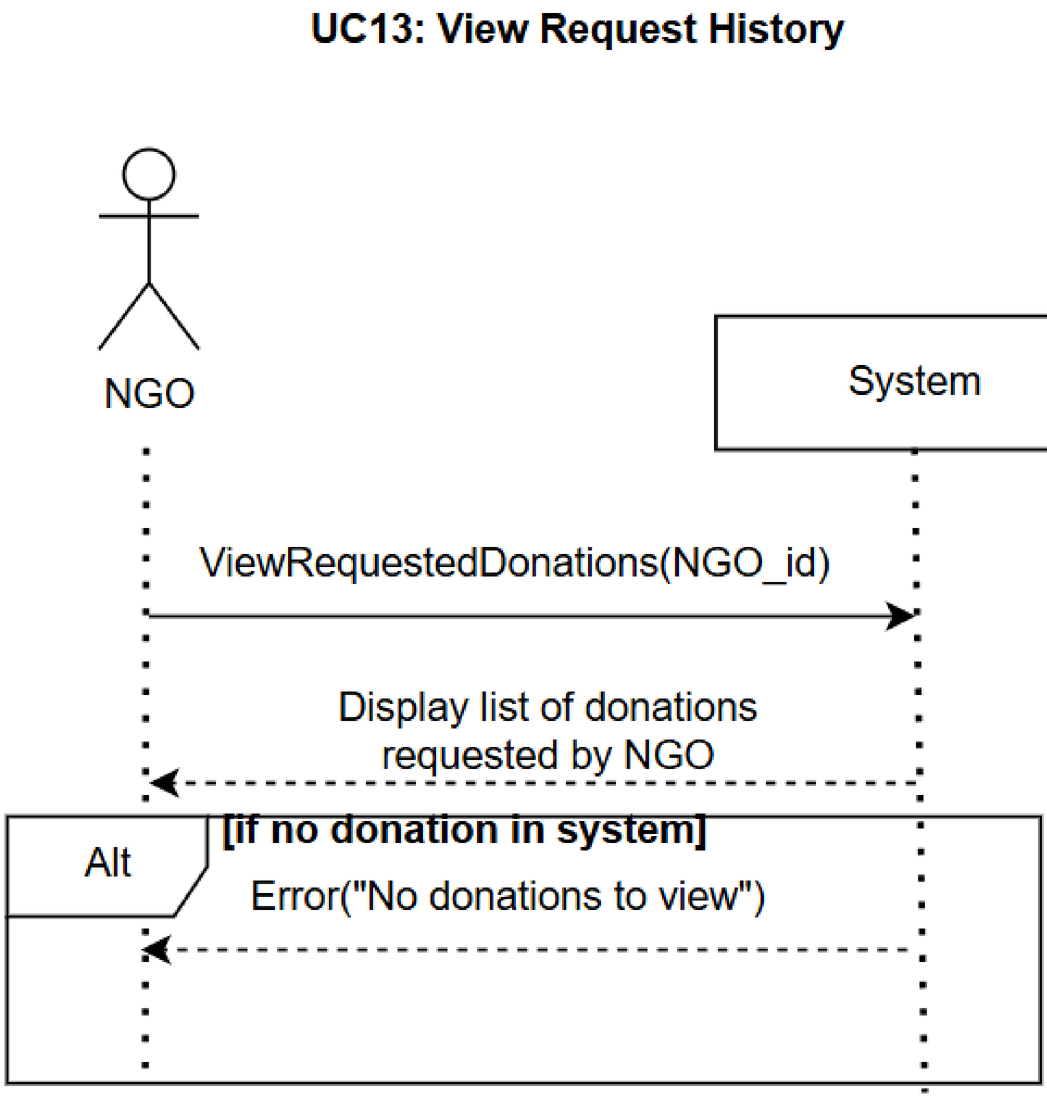


Figure 2.15: View Request History

UC-14: Request Available Donation

Figure 2.16 shows the system sequence diagram of Request Available Donation, NGO sends request to request donation that are currently available to the system, and the system responds by creating the request and if request was already made by the NGO, then the system responds with the message of error.

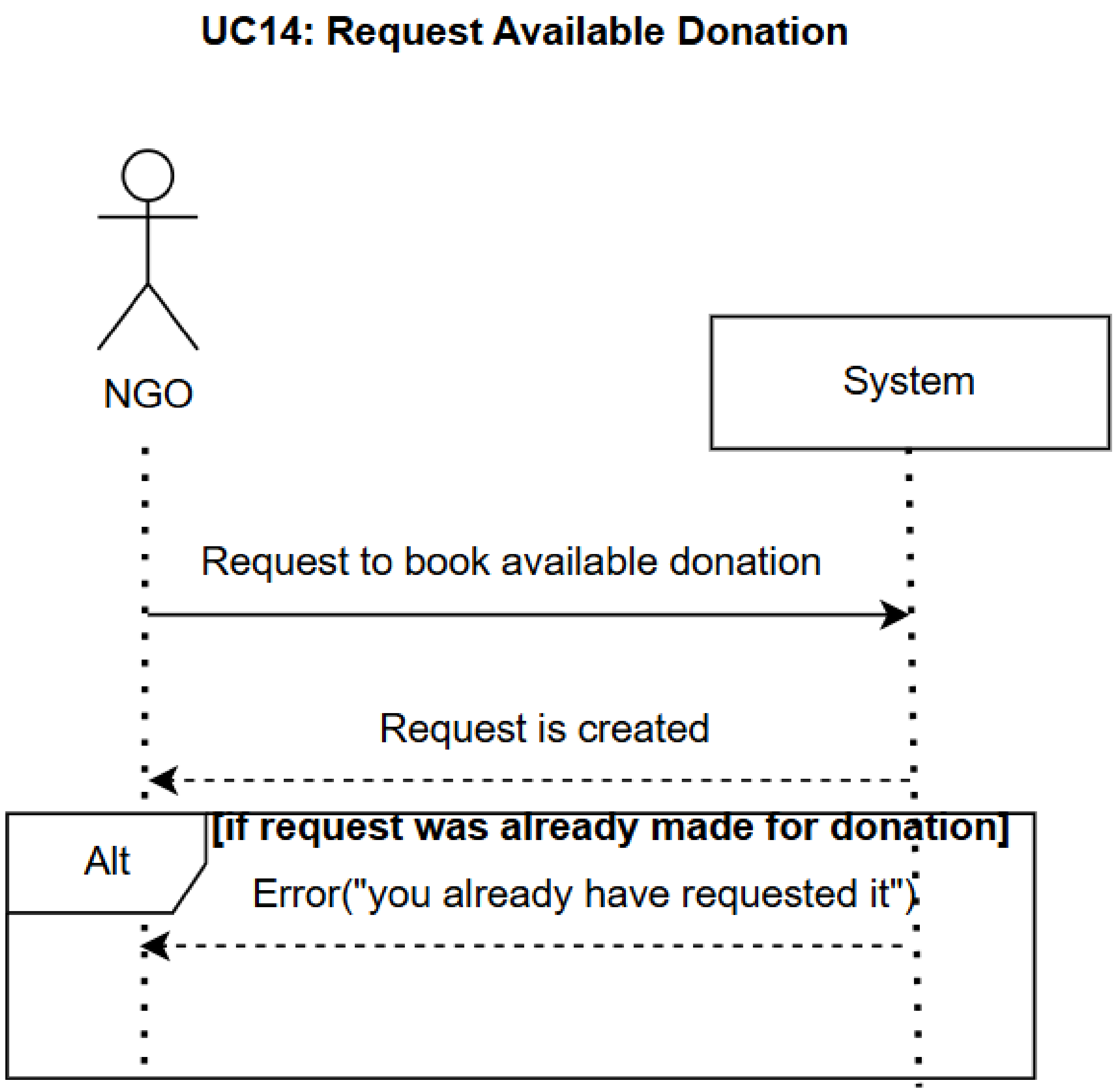


Figure 2.16: Request Available Donation

UC-15: Mark Booked Donations as Collected

Figure 2.17 shows the system sequence diagram of Mark Booked Donations as Collected, NGO views all the donations that NGO booked to collect and then for a booked donation make request to mark it collected after collecting it from pickup location, and the system responds by marking that donation as collected

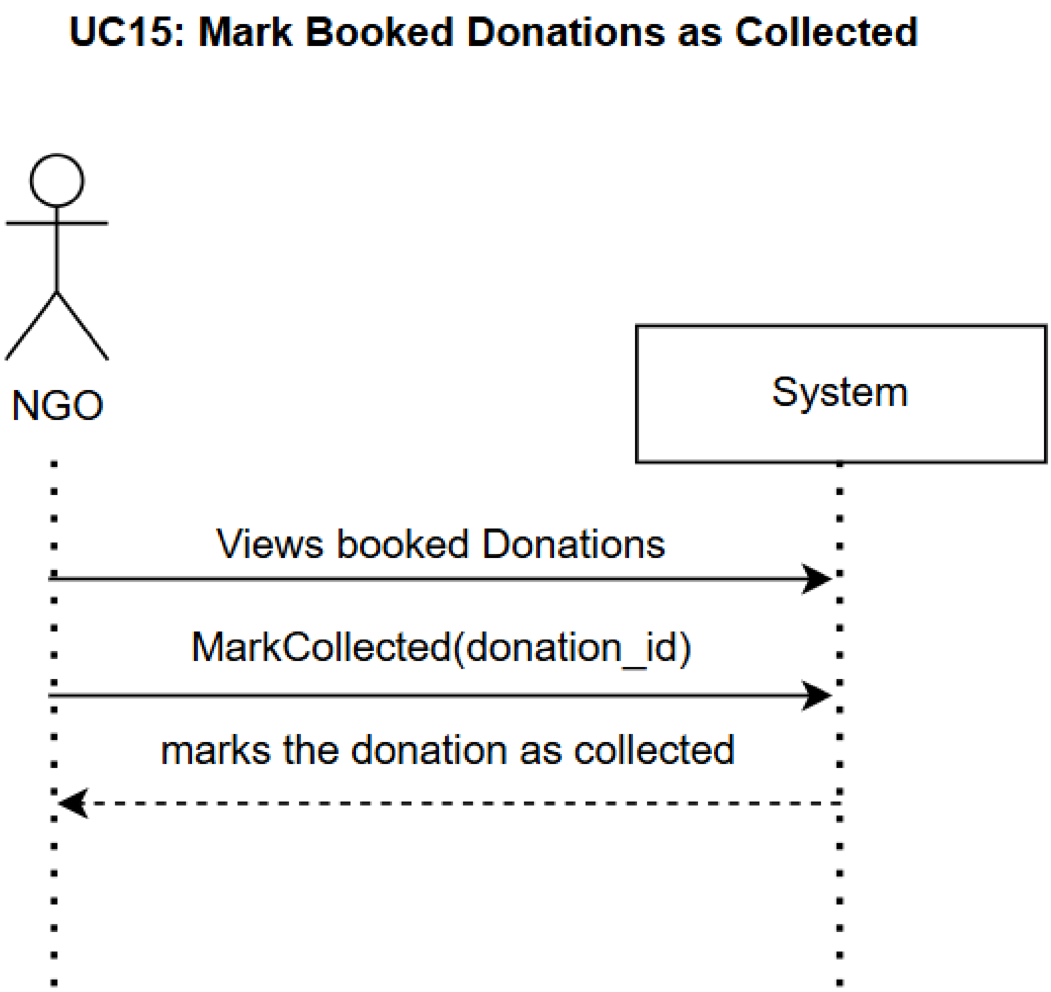


Figure 2.17: Mark Booked Donations as Collected

UC-16: View NGO Donation history

Figure 2.18 shows the system sequence diagram of View NGO Donation history, NGO make request view history of all the donations requested, booked or collected by NGO to the system, and the system responds with the list of donations and then NGO applies filter and the system responds with the updated list of donations and if no donation to show is available, then the system responds with the message of error.

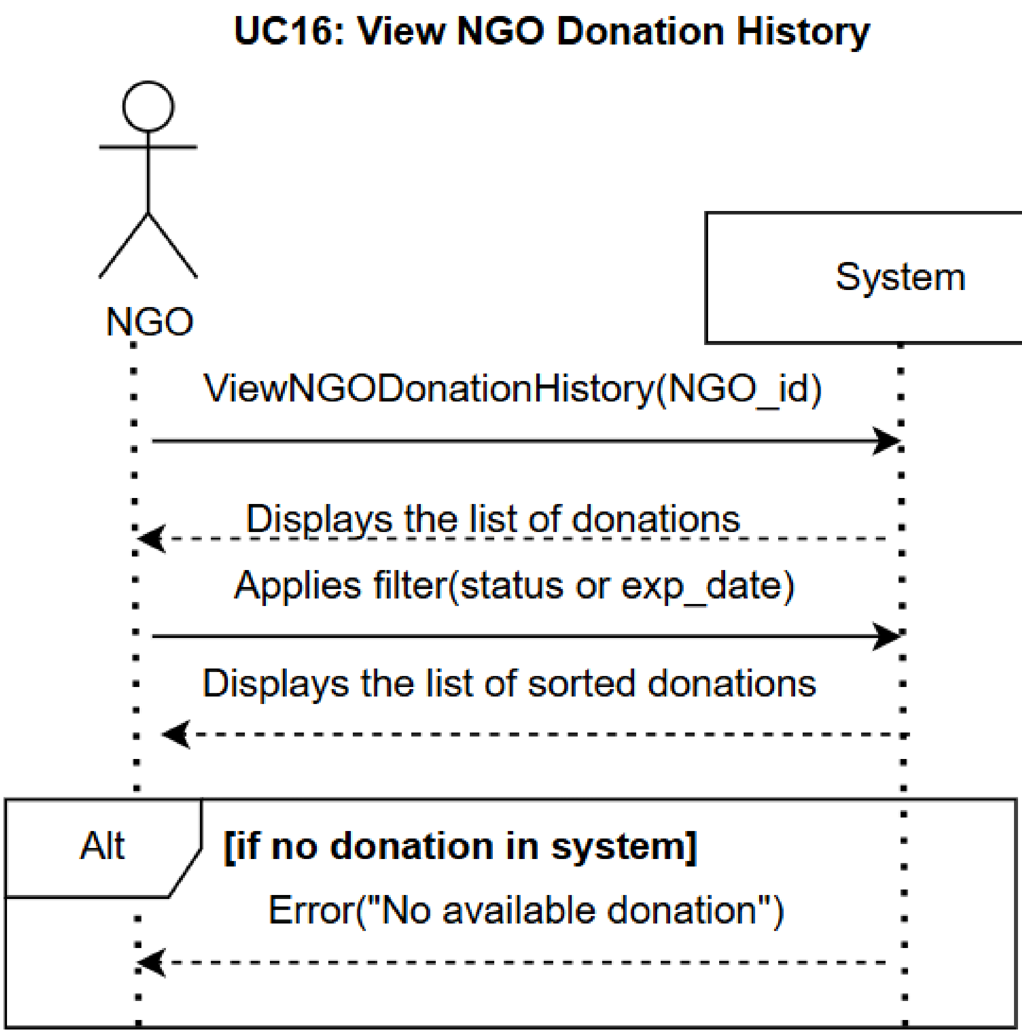


Figure 2.18: View NGO Donation History

UC-17: View NGO Profile

Figure 2.19 shows the system sequence diagram of view NGO profile for the NGO sending view NGO profile request to the system, and the system responds with the NGO of donor.

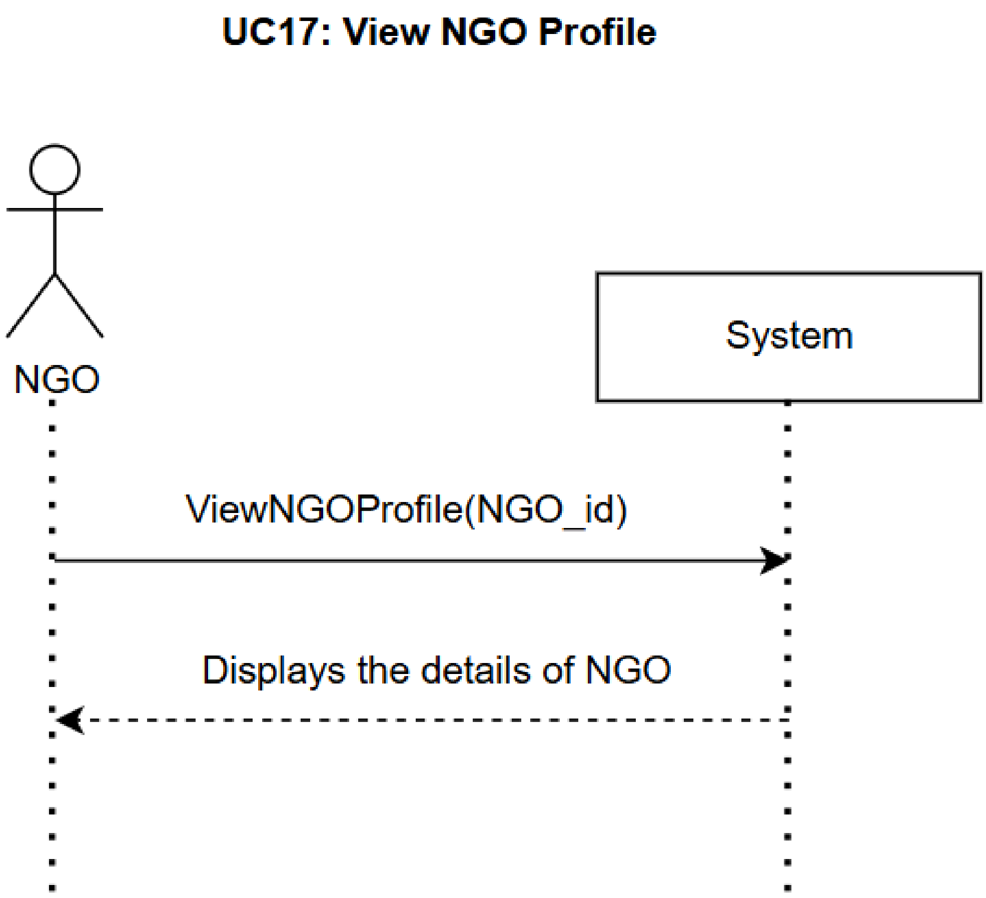


Figure 2.19: View NGO Profile

UC-18: Edit NGO Profile

Figure 2.20 shows the system sequence diagram of Edit NGO Profile, NGO make Edit NGO Profile request to the system, and the system returns the details of NGO, then NGO submits the new information and system responds with confirmation message and if information provided by NGO is incomplete, then the system responds with the message of incomplete detail and if information provided by NGO is invalid, then the system responds with the message of invalid detail and if information provided by donor is not editable, then the system responds with the message of it.

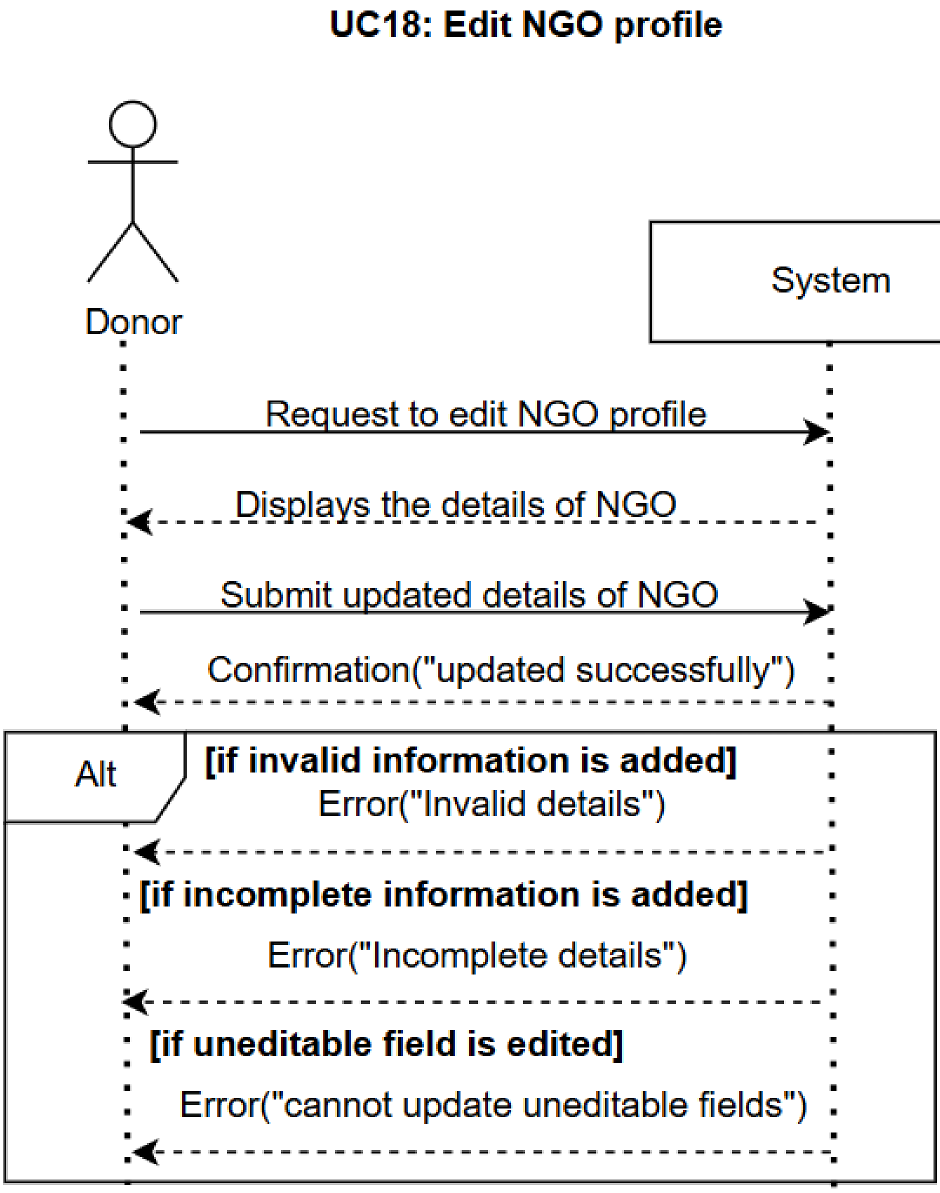


Figure 2.20: Edit NGO Profile

UC-19: Verify NGOs Signup

Figure 2.21 shows the system sequence diagram of Verify NGOs Signup, Admin make request to view all the pending verification requests of NGOs of Signup so that there account is verified to login, and the system returns the list of NGOs, then Admin make view NGO request to view the details of NGO and system responds with detail of NGOs and then NGO approves or rejects the NGO and system responds by setting NGOs status approved or rejected.

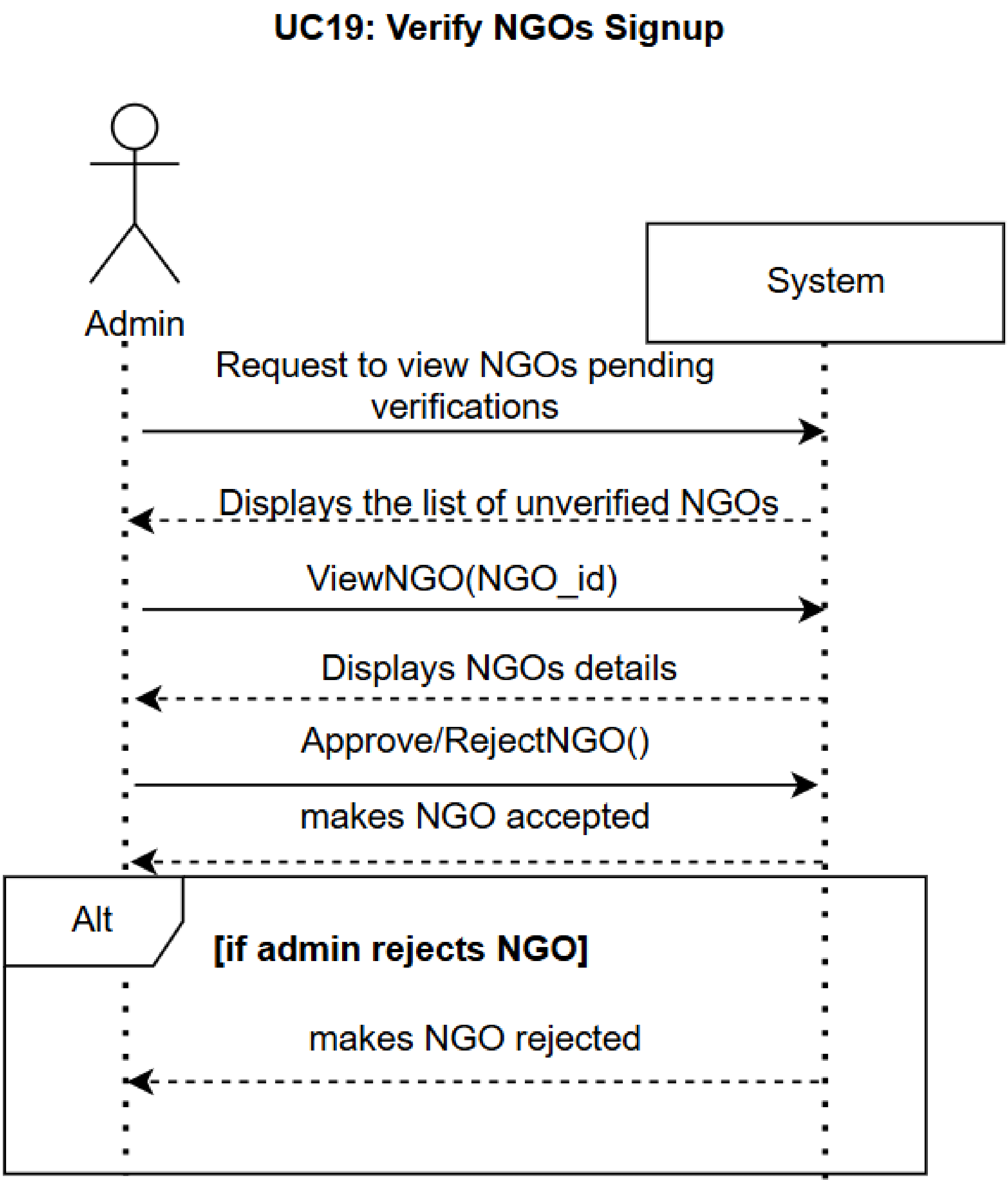


Figure 2.21: Verify NGOs Signup

UC-20: Delete User

Figure 2.22 shows the system sequence diagram of Delete User, Admin make request to view all the users of the system donor and NGOs, and the system responds with the list of Users and then Admin sends request to delete a user and the system responds with confirmation message.

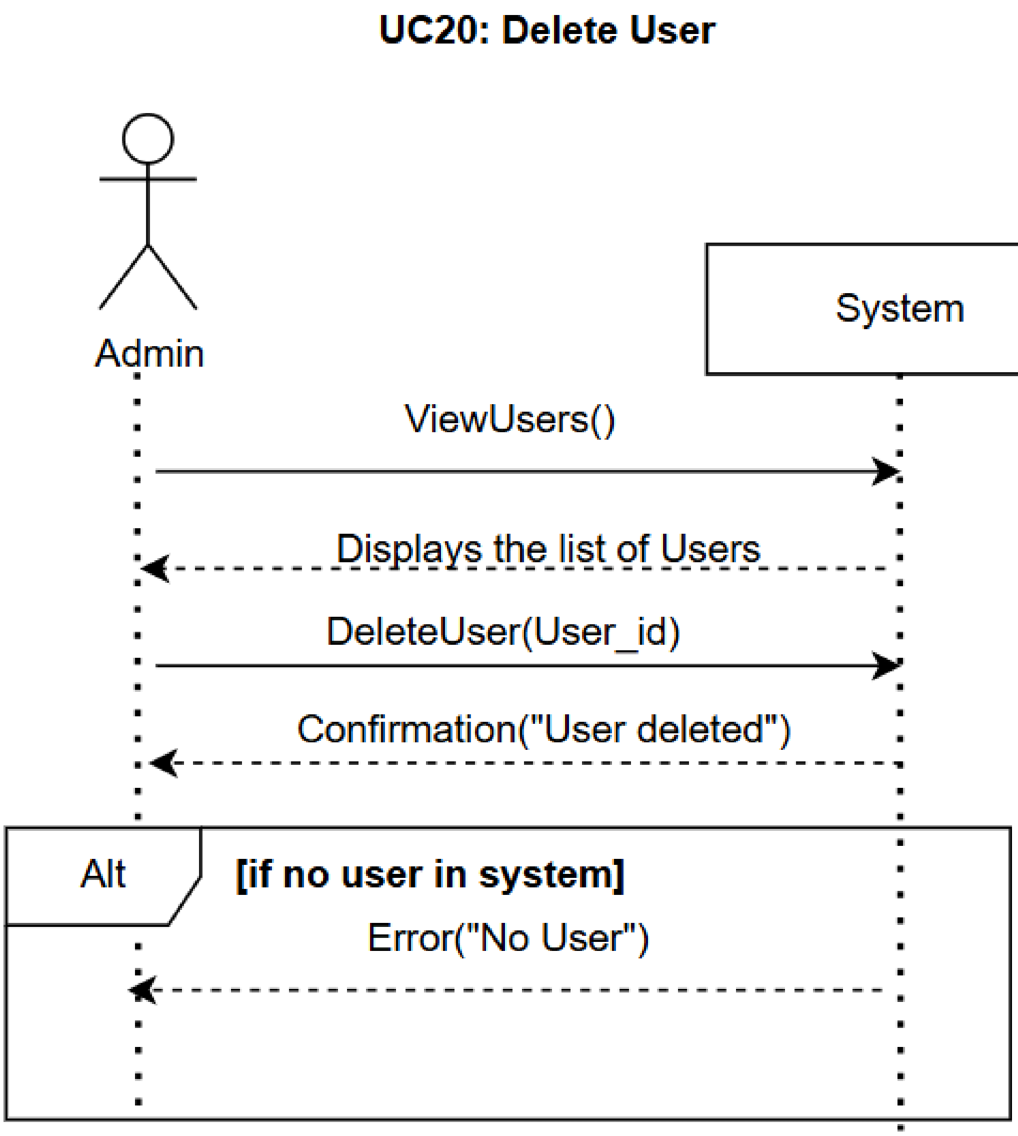


Figure 2.22: Delete User

UC-21: View User List

Figure 2.23 shows the system sequence diagram of View User, Admin make request to view all the users of the system donor and NGOs, and the system responds with the list of Users and if no user to show is available, then the system responds with the message of error.

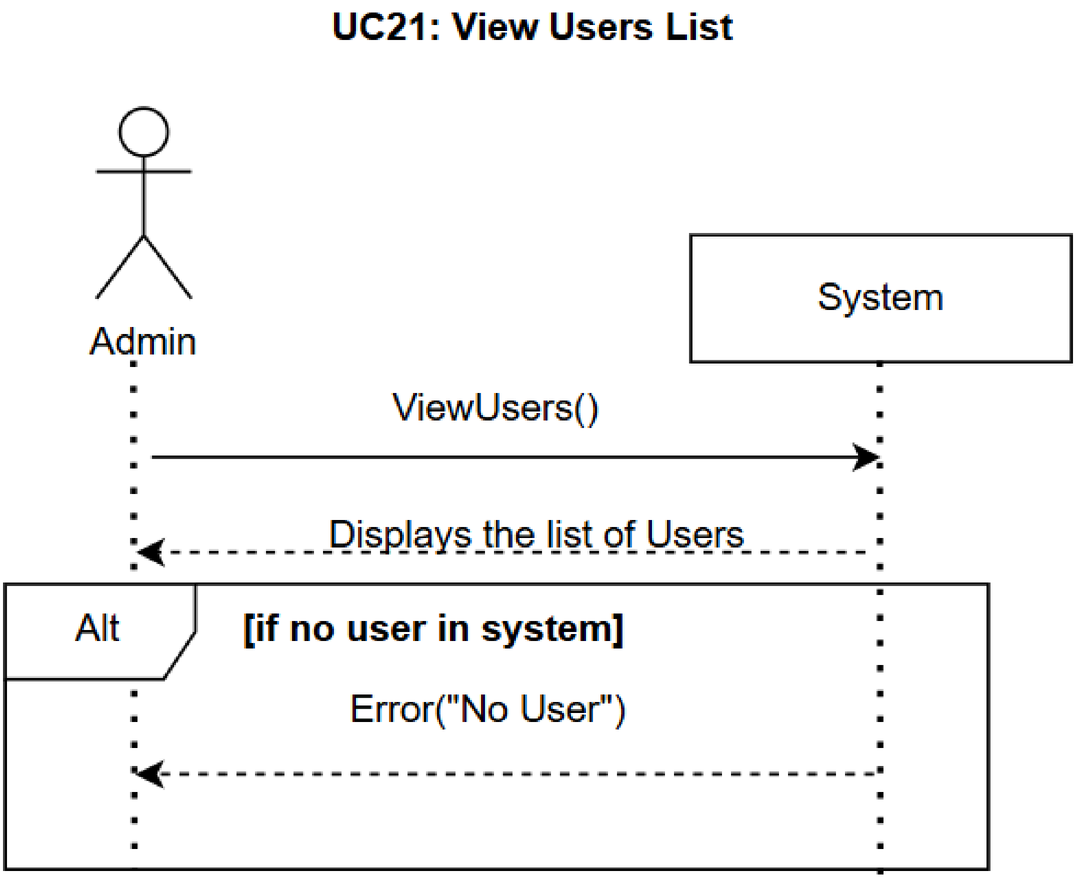


Figure 2.23: View Users List

UC-22: View All Donations

Figure 2.24 shows the system sequence diagram of View All Donation, Admin make request to view all the donation of the system, and the system responds with the list of donors and if no donation to show is available, then the system responds with the message of error.

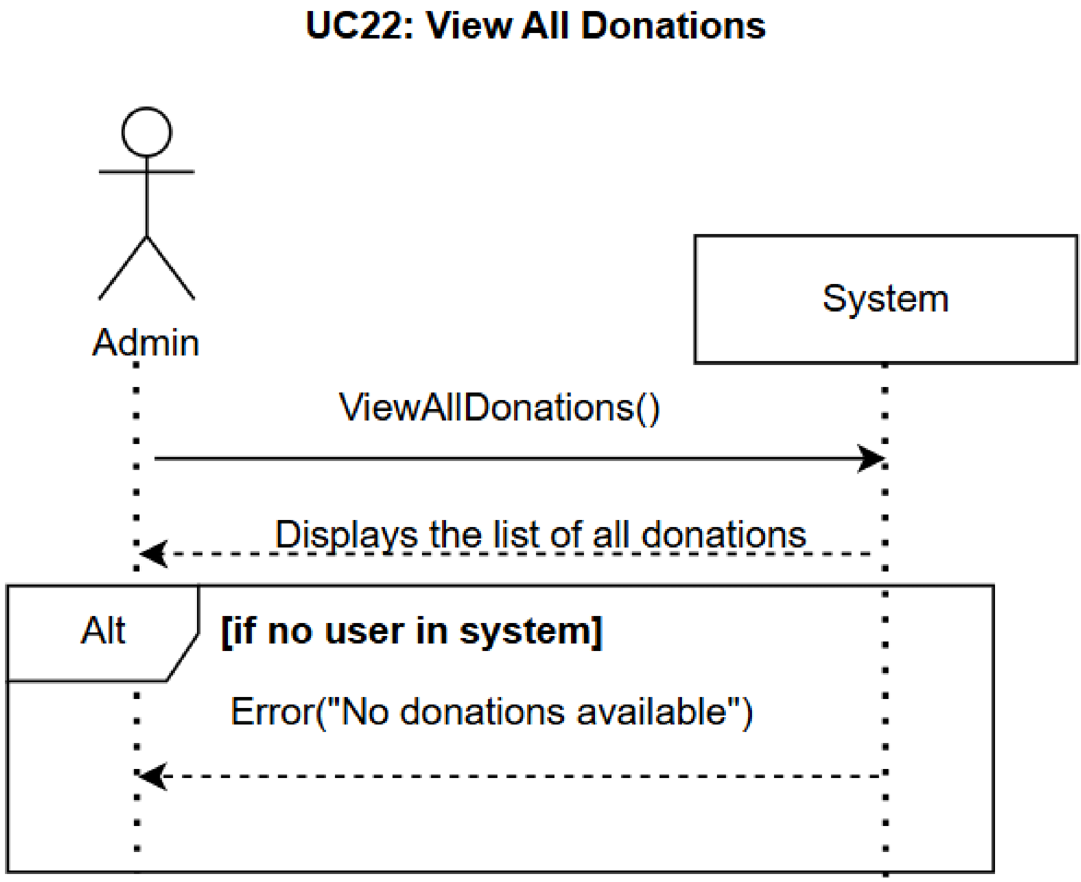


Figure 2.24: View all donations

UC-23: View NGO requests

Figure 2.25 shows the system sequence diagram of View NGOs Request, Admin make request view all request of NGOs for booking a donations, and the system responds with the list of donation requests of NGOs and if no donation to show is available, then the system responds with the message of error.

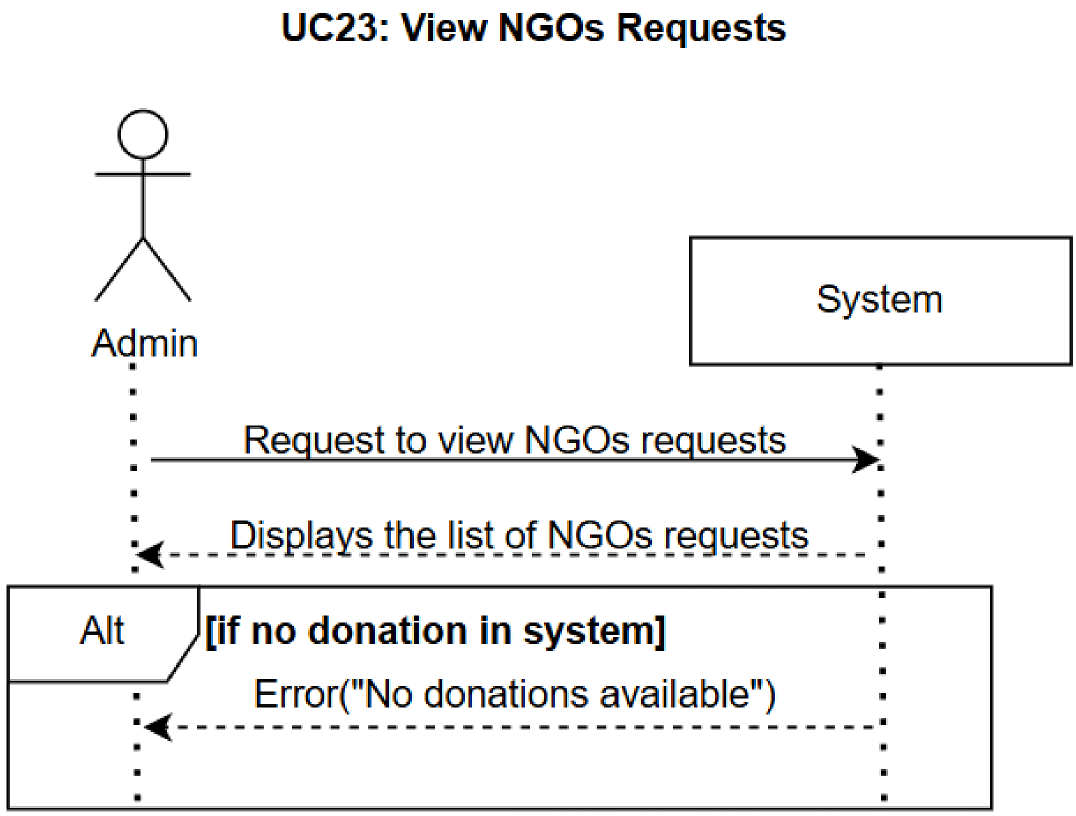


Figure 2.25: View NGO Requests

UC-24: Accept Requested Donation

Figure 2.26 shows the system sequence diagram of Accept Requested Donation, Admin make request to view all request of NGOs for booking a donation, and the system responds with the list of donation requests of NGOs then admin approves any one NGO for a donation and system responds by booking that donation that NGO and auto rejecting all other NGOs requests and if Ngo does not have capacity to accept the donation , then the system responds with the message of error.

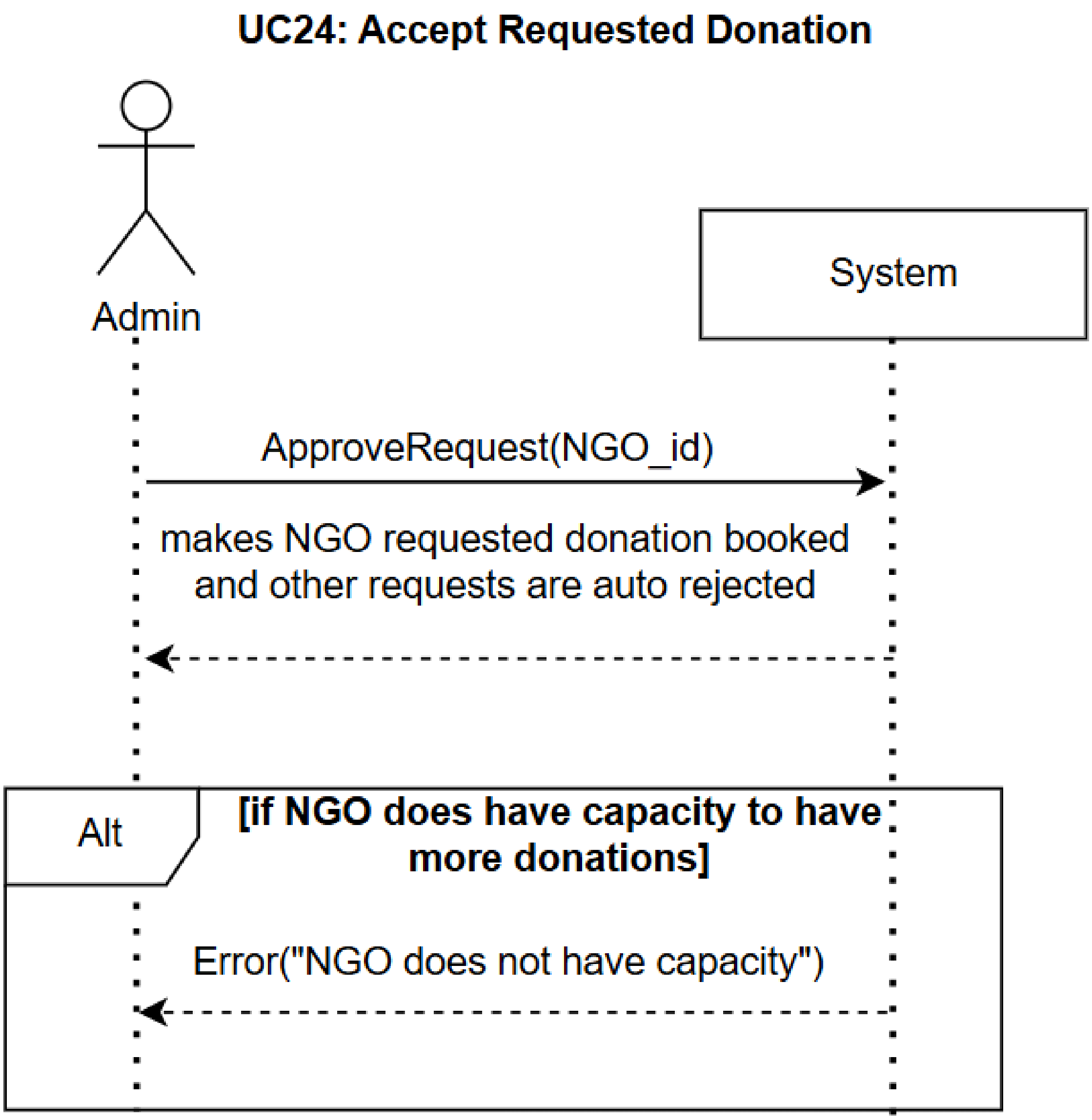


Figure 2.26: Accept Requested Donation

UC-25: Manage Donation Status

Figure 2.27 shows the system sequence diagram of Manage Donation Status, Admin make request to view a donation in the system, and the system responds with donation with its details and status then Admin request to edit the status of donation with available status with giving the updating status and system responds by confirmation message and if the previous status of donation was available , then the system responds with the message of error.

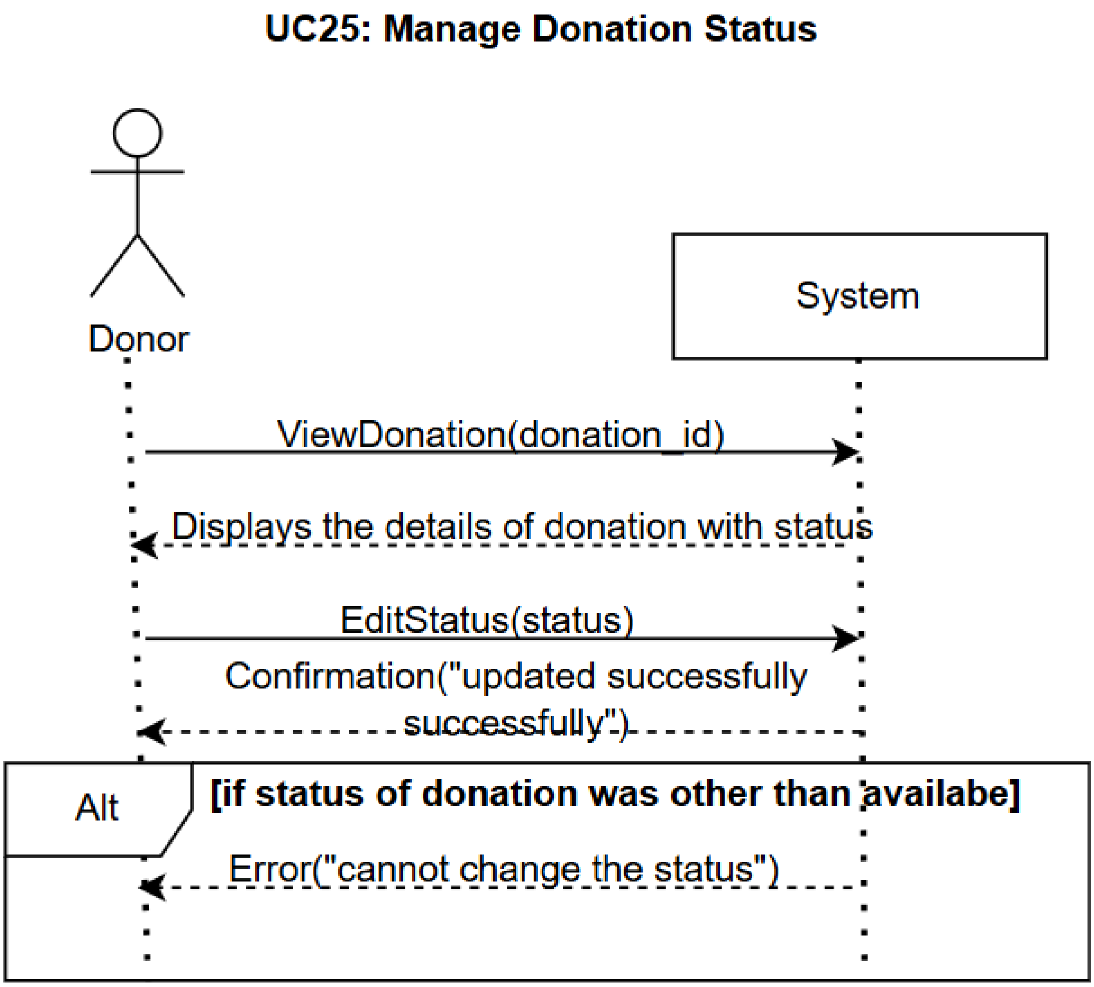


Figure 2.27: Manage Donation Status

2.7 Domain Model

Figure 2.28 shows the domain model of Local Food Distribution System, domain model shows real world entities relationship, and in our system the domain model entities are User that is abstract inherited by Donor, NGO and Admin, other domain entities are Notifications, Donations , Donation request that represent all the requests of donation of NGO and NGOVerificationReview that represent Ngo registration requests. Then their attributes are mentioned in the diagram and what will be the relationships between the entities and also the cardinalities.

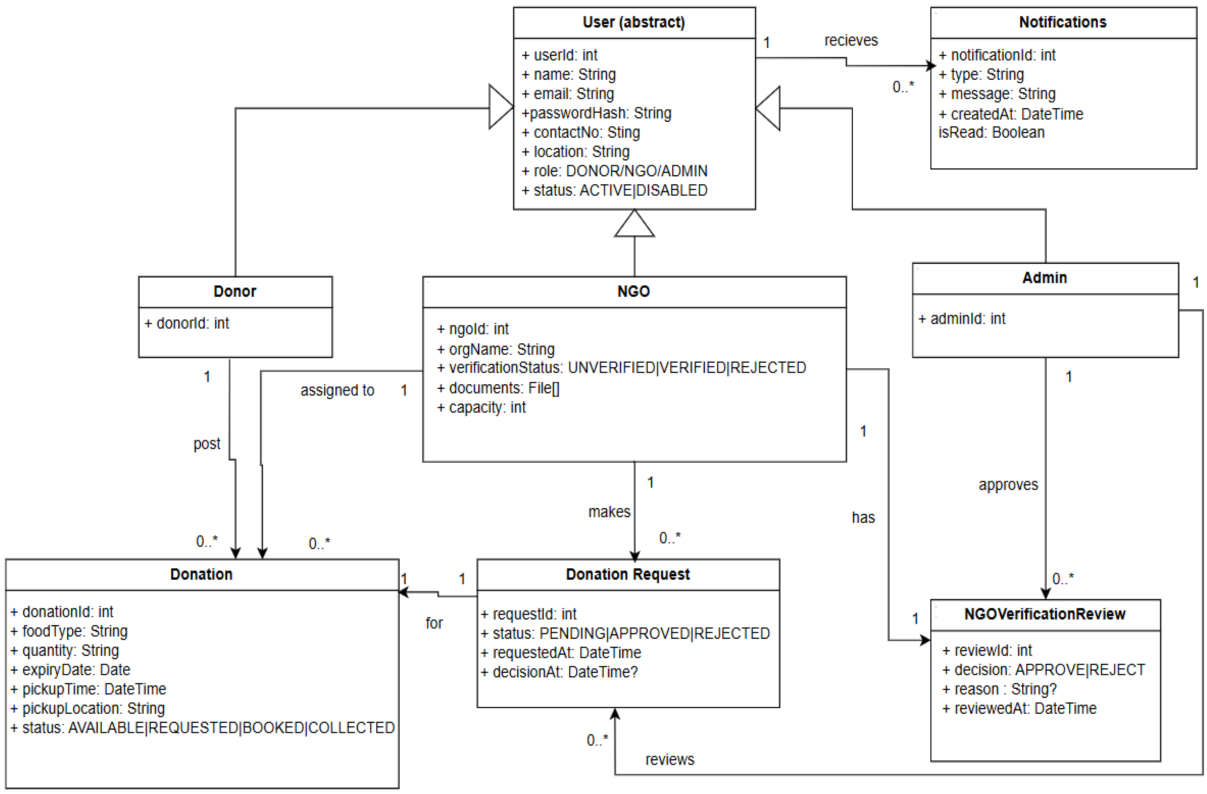


Figure 2.28: Domain Model

3. Software Design Description(SDD)

3.1 Introduction

This chapter provides an overview of system design and system architecture of Local Food Distribution System. It presents architectural diagrams describing and showing the structure of the system. It also shows the compliance with the functional and technical specifications.

3.1.1 Design Overview

The design of the Local Food Distribution System follows modular architecture to ensure proper scalability, maintenance and clarity of responsibilities among the system. The Local Food Distribution System is designed as a web-based application that connects donors, NGOs and administrators through a centralized platform to manage food donation.

3.2 Requirements Traceability Matrix (RTM)

Table 3.1 shows the requirement traceability matrix, requirement traceability matrix establishes a clear and transparent relationship between the requirements of the system and their interfaces whether they exist or not and their respective test cases. This table also show the list of all requirement ids and their names with the interfaces of the requirements and their test cases.

Requirement ID	Requirement name	Interface	Test Case
UC-1	Login	Login Screen	TC-1
UC-2	Logout	Logout	TC-1
UC-3	View Notifications	View Notifications Page	TC-2
UC-4	Donor Signup	Donor Signup Form	TC-3
UC-5	Post Donation	Post Donation Form	TC-4
UC-6	Edit Available Donation	Edit Available Donation Form	TC-4
UC-7	Delete Available Donation	Donor Donation List	TC-4
UC-8	View Donor Donation history	Donor Donation History Screen	TC-2
UC-9	View Donor Profile	Donor Profile Screen	TC-5
UC-10	Manage Donor Profile	Donor Edit Profile Screen	TC-5
UC-11	NGO Signup	NGO Signup Screen	TC-3
UC-12	View Available Donations	NGO Available Donations Screen	TC-2

UC-13	View request History	NGO Request History Screen	TC-2
UC-14	Request Available Donation	NGO Donation Detail	TC-6
UC-15	Mark Booked Donations as Collected	NGO Booked Donations Screen	TC-6
UC-16	View NGO Donation history	NGO Donation History Screen	TC-2
UC-17	View NGO Profile	NGO Profile Screen	TC-5
UC-18	Manage NGO Profile	NGO Edit Profile Screen	TC-5
UC-19	Verify NGOs Signup	Admin Pending Verifications Screen	TC-7
UC-20	Delete User	Admin User Management Screen	TC-7
UC-21	View User List	Admin User Management Screen	TC-7
UC-22	View all donations	Admin All Donations Screen	TC-2
UC-23	View NGO requests	Admin Donation Requests Screen	TC-2
UC-24	Accept requested donation	Admin Donation Requests Screen	TC-8
UC-25	Manage Donation Status	Admin Donation Detail Screen	TC-8

Table 3.1: Requirements Traceability Matrix

3.3 Class Diagram

Figure 3.1 show the class diagram of Local Food Distribution System, class diagram is the UML diagram that displays and describes the static structure of the system. The figure below does the same and is describing the static structure of the system, the attributes in the system and all the methods in the system, the class diagram given below also shows all the classes in the system and the relationships in the system. The classes in our system are User that is abstract inherited by Donor, NGO and Admin, other classes are Notifications, Donations , Donation request that represent all the requests of donation of NGO and NGOVerificationReview that represent Ngo registration requests.

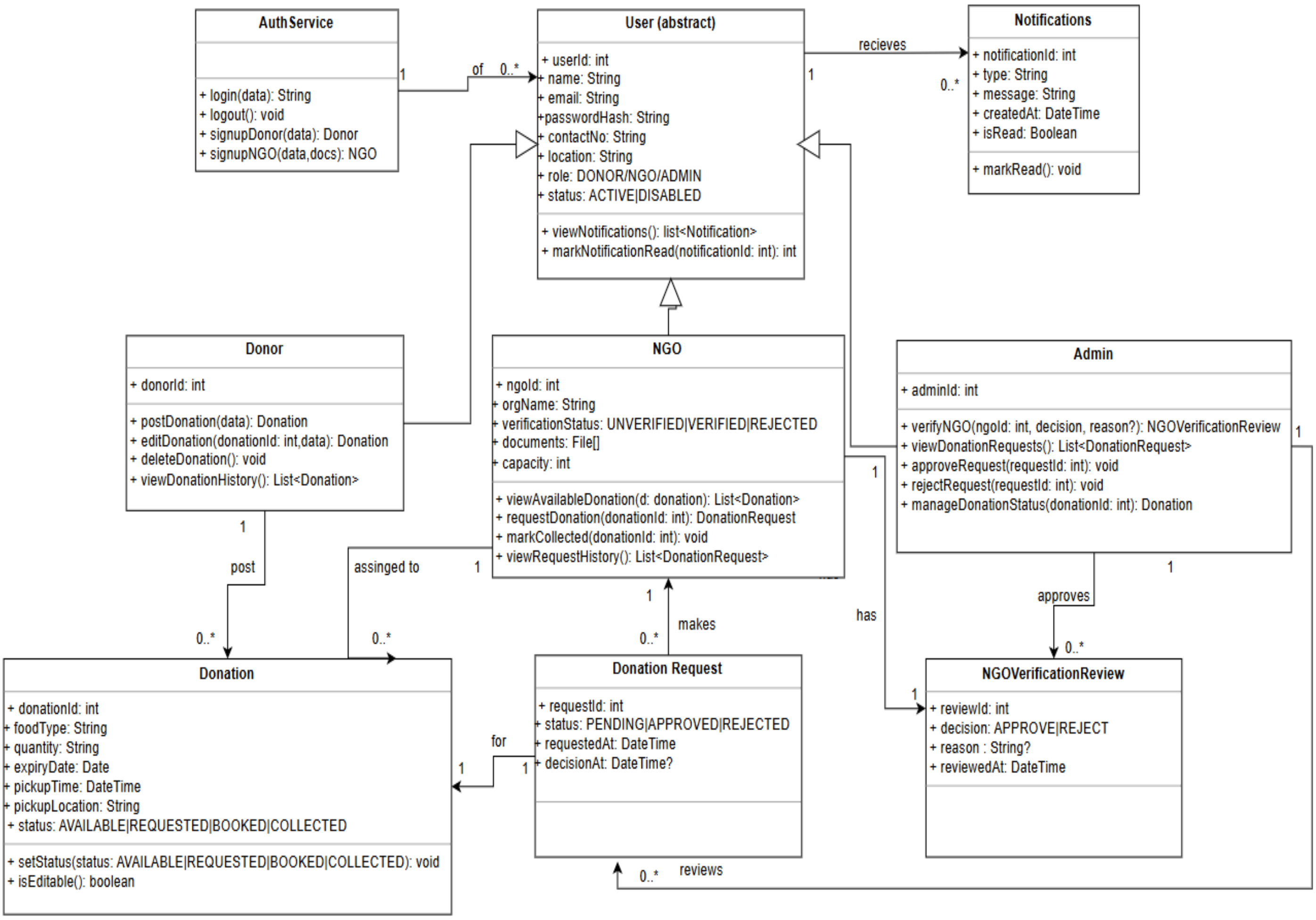


Figure 3.1: Class Diagram

3.4 Entity Relationship Diagram (ERD)

Figure 3.2 shows the entity relationship diagram, ERD is a visual tool used to design databases, in database design to model and design ERD describes the data within the system. This entity relationship diagram describes the logical structure of databases by showing their entities, attributes and relationships between them, the primary and foreign keys in the system, where primary keys are the unique identifier and foreign keys are the reference to another entity's primary keys. All the relationships in the system clearly defined with cardinality

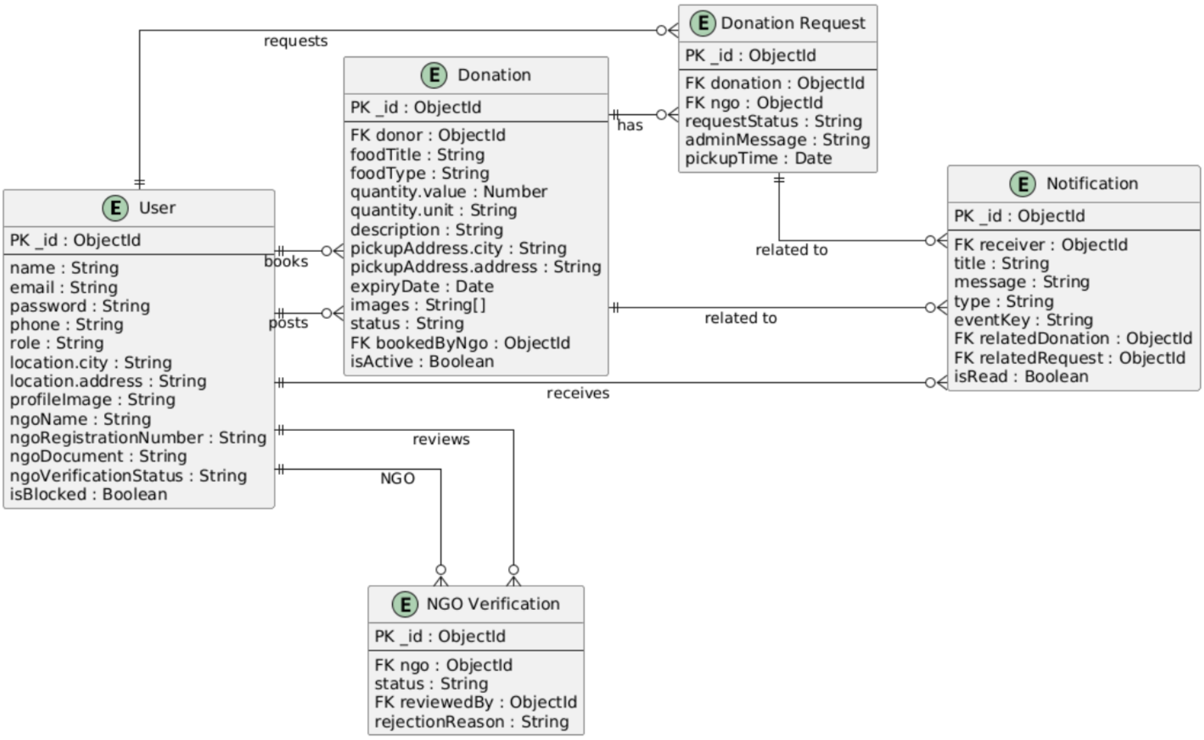


Figure 3.2: Entity Relationship Diagram

4. Software Implementation(SI)

4.1 Introduction

This chapter briefly discusses how all the software design made in previous chapter was converted into the Local Food Distribution System that made it a web application that can be used by users and facilitates them.

4.2 Overview of the System

Purpose:

The Local Food Charity System is web-based application that solves the problem of saving surplus food from getting wasted. It basically connects NGOs and Donors to distribute and collect food transparently under the monitoring of Admin.

Primary Roles:

1. Donor: In the system, donor donates the food to save food from getting wasted. For this he register and log in, post new donations, edit or delete available donations, view their donation history, manage their profiles and receive notifications regarding donation activities.
2. Seller: In the System, NGO accepts the donation by providing their NGO information and verification documents. After approval by the Admin, NGO can login, view available donations, submit donation requests, view request history, manage booked donations, mark donations as collected, manage its profile and receive notifications.
3. Admin: In the system, admin manages and monitors the system by verifying NGO registrations, managing users, viewing donations, reviewing donation requests from NGOs, approving suitable NGO for a donation, manages donation statuses and monitoring activities within the platform.

Core Modules:

The core modules of the system are Authentication Module, Donor Module, NGO Module, Admin Module and Notification Module.

Key Flow:

Donor Registration/Login → Post Donation → Donation Available → NGO Registration and Verification → NGO Views Available Donation → NGO Submits Request → Admin Reviews Requests → One NGO Request Approved → Other Requests Rejected → Donation Booked → NGO Collects Donation → Donation Marked as Collected

4.3 Development Environment and Technologies

Here we are going to discuss all the technologies and environment setup in which project was build ,run and deployed.

4.3.1 MERN Stack Development

1. MongoDB- Storage and management of application data
2. Express.js- Development of backend RESTful APIs
3. React.js- Development of the frontend user interface
4. Node.js- Runtime environment for backend development

4.3.2 Reason for MERN Stack

1. Single Programming Language- One language can be used for both frontend and backend avoiding the hassle of switching languages.
2. Modularity and Maintainability- MERN supports separating the application into frontend, backend and database components, making the implementation easier.
3. REST API Development- Node.js and Express.js provide a suitable environment for developing RESTful APIs.
4. Flexible Data Storage- MongoDB provides document-based storage suitable for managing system information.

4.3.3 Development Environment

1. Visual Studio Code - source code editor
2. Node.js runtime environment
3. npm - package/dependency management
4. Google Chrome - running and testing the web application
5. Postman - testing RESTful APIs
6. MongoDB Atlas - cloud database environment
7. Git/GitHub- version control

4.3.4 Frontend Technologies and Libraries

1. React.js
2. JavaScript
3. HTML
4. CSS
5. Tailwind CSS
6. React Router
7. Axios

4.3.5 Backend Technologies and Libraries

1. Node.js
2. Express.js
3. RESTful APIs
4. JSON Web Token (JWT)
5. bcrypt
6. Mongoose
7. Rate limiter
8. Multer
9. Node mailer
10. Cookie parser
11. Express validator

12. Cors
13. Dotenv

4.3.6 Database and Cloud Services

1. Cloudinary
2. MongoDB Atlas

4.3.7 Deployment Environment

1. Vercel
2. Render

4.4 System Architecture Implementation

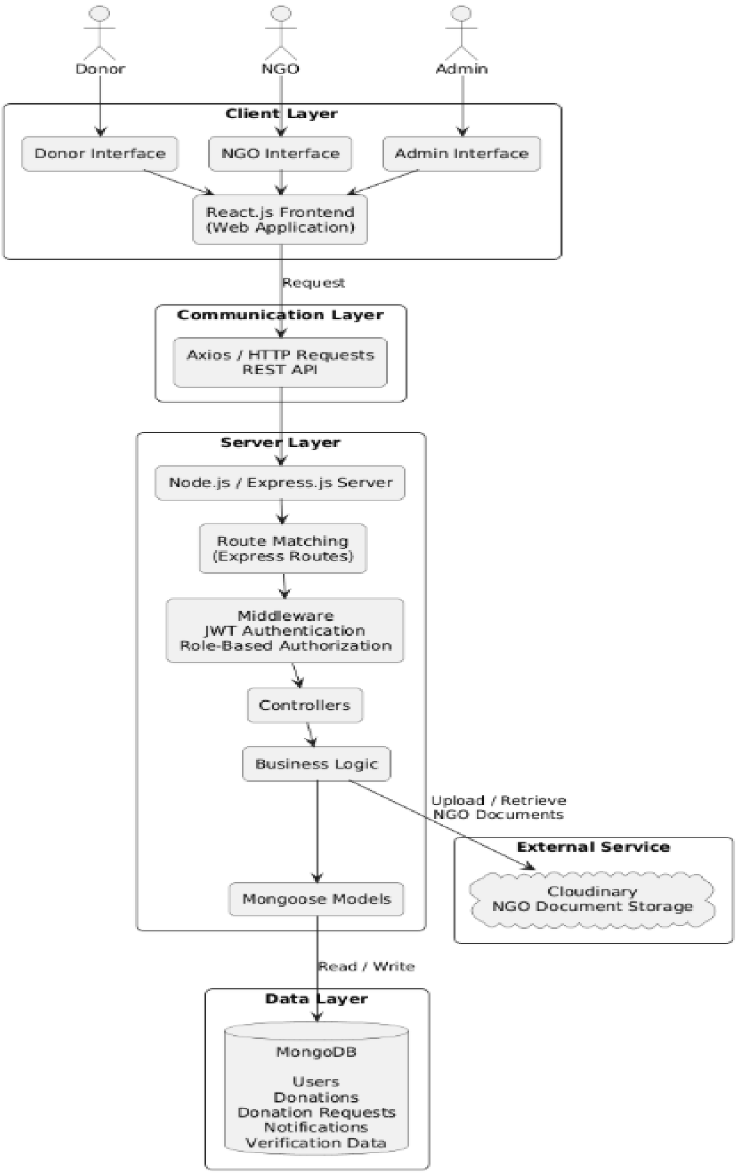


Figure 4.1: Architecture Implementation Diagram

Figure 4.1 shows the architecture that is implemented in the software. First, it shows that all users whether its donor, NGO, or admin they interact with system only through their respective interface and that interface sends the request to frontend system then frontend system through REST API methods sends request to the node.js and node.js passes it to express.js. From there it matches the respective route and middleware for that routes are called, middlewares then call the next function in which business logic is written and at the end server layer interacts with data layers and cloudinary.

4.5 Frontend Development

4.5.1 Technology Stack

- **Framework**: React.js
- **Build Tool:** Vite
- **Programming Language:** JavaScript
- **State Management:** Context API and Custom Hooks
- **UI/Styling:** CSS and Tailwind CSS
- **Routing:** Frontend route-based navigation

4.5.2 Project Structure

```
Local Food Distribution System/
└── frontend/
    ├── public/
    │
    ├── src/
    │   ├── components/
    │   ├── config/
    │   ├── constants/
    │   ├── context/
    │   ├── hooks/
    │   ├── layouts/
    │   ├── pages/
    │   ├── routes/
    │   ├── sections/
    │   ├── services/
    │   ├── styles/
    │   ├── utils/
    │   ├── App.jsx
    │   └── main.jsx
    │
    ├── .env
    └── package.json
```

4.5.3 Frontend Architecture

Main.jsx is the entry point and it renders the app and loads App.jsx, then it manages the authentication and routing, then it displays the appropriate pages through the layouts also using reusable components and communicates with backend with services and RESR APIs.

main.jsx
↓
App.jsx
↓
AuthContext and Router
↓
Public/Protected Routes
↓
Layouts
↓
Pages
↓
Reusable Components
↓
Custom Hooks and Services
↓
Backend REST APIs

4.5.4 Key Features Implemented

The frontend basically handles this and provides interface for them

Authentication Interface

- It gives secure login, registration, protected user access and logout.
- It also gives OTP verification.
- It redirects the user accordingly.

Donor Interface

- First, it provides the donor dashboard.
- He can create, delete, edit, and view donations.
- Track the donation and request statuses.
- Donor interface also displays relative notifications, histories of donations and statistics.
- He can view and manage his profile.

NGO Interface

- First, it provides an NGO dashboard.
- He can view all available donations and donations with other statuses
- Track the donation and request statuses.
- NGO interface also displays relative notifications, histories of donations collected and statistics.
- He can view and manage his profile.

Admin Interface

- First of all it provides the Admin dashboard.
- He user can create , delete, edit, and view donations with other statuses.
- Track the donation and request statuses and manipulate them.
- He can view and manage his profiles of users.
- Admin interface also displays relative notifications and statistics.

Donation Management

- Donors can post donations, edit or delete available donations, and view donation history.
- Donation can be tracked.

Donation Request Management:

- Allows NGO to submit documentation
- Admin approve or reject it
- Allow donor , NGO and Admin to track the status of request.

Notification Interface:

- Users can view latest notifications of system activities.

4.6 Backend Development

4.6.1 Technology Stack

- **Runtime:** Node.js
- **Framework:** Express.js
- **Database:** MongoDB / MongoDB Atlas
- **Authentication:** JSON Web Token (JWT)
- **Architecture:** Routes, Controllers, Models, Middleware, and Services
- **Cloud Storage:** Cloudinary for NGO verification documents
- **Configuration:** Environment-based configuration using .env

4.6.2 Project Structure

```
Local Food Distribution System/
│
├── backend/
  ├── config/
  ├── constants/
  ├── controllers/
  ├── middleware/
  ├── models/
  ├── routes/
  ├── services/
  ├── utils/
  ├── .env
  ├── server.js
  └── package.json
```

4.6.3 Backend Architecture

The client sends the request, it goes to express server, then respective route is match and their middleware are then used to make routes protected and the next business logic is executed in controller for which to access data model is used to collaborate with MongoDB to give JSON response.

Client Request
↓
Express Server
↓
Route
↓
Middleware
↓
Controller
↓
Model
↓
MongoDB
↓
JSON Response

4.6.4 API Implementation

This table gives he module in the system their respective endpoints and methods of that endpoint with that the table also gives the purpose of those endpoints and their authentication.

Module	Method	Endpoint	Purpose	Authentication
Authentication	POST	/api/auth/login	Authenticate user	Public
Authentication	POST	/api/auth/logout	Log out user	Authenticated
Authentication	GET	/api/auth/profile	View user profile	All roles
Donor	POST	/api/donations	Create donation	Donor
Donor	PUT	/api/donations/:id	Edit donation	Donor

Module	Method	Endpoint	Purpose	Authentication
Donor	DELETE	/api/donations/:id	Delete donation	Donor
NGO	GET	/api/ngo/donations/available	View available donations	Approved NGO
NGO	POST	/api/ngo/donations/:id/request	Request a donation	Approved NGO
NGO	PUT	/api/ngo/requests/:id/cancel	Cancel a request	Approved NGO
NGO	PUT	/api/ngo/donations/:id/collect	Mark donation as collected	Approved NGO
Admin	GET	/api/admin/requests	View donation requests	Admin
Admin	PUT	/api/admin/requests/:id/approve	Approve a request	Admin
Admin	PUT	/api/admin/requests/:id/reject	Reject a request	Admin
Admin	PUT	/api/admin/status/:id	Change donation status	Admin
Notifications	GET	/api/notifications	View notifications	All roles
Notifications	PUT	/api/notifications/:id/read	Mark notification as read	All roles

Table 4.1: API Implementation

4.6.5 Core Business Logic

1. **User Authentication:** Handles user registration, login, logout, and authenticated access.
2. **Role-Based Authorization:** Only allow appropriate users for the system operations based on role.
3. **Donation Management:** Handles creation, viewing, updating, deletion, and status management of donations.
4. **Donation Request Management:** Allows verified NGOs.
5. **Donation Lifecycle:** backend implements the donation lifecycle as Created → Available → Requested → Booked → Collected
6. **Donation Lifecycle:** backend implements the donation lifecycle as Created → Pending → Approved/Rejected/Cancelled

Its complete example is

Donor posts donation
↓
Donation becomes available
↓
NGO submits request
↓
Request becomes pending
↓
Admin approves one request
↓
Donation becomes booked
↓
Other requests are rejected
↓
NGO collects donation
↓
Donation and request become collected

4.6.6 Security Implementation

1. **JWT Authentication:** Authentication tokens are used to protect secured API operations.
2. **Role-Based Authorization:** Only allow appropriate users for the system operations based on role.
3. **Password Protection:** Password are secured from being stolen by using bcrypt.
4. **Environment Variables:** Sensitive configuration are stored in .env

4.7 Database Implementation

4.7.1 Database Technology

- **DBMS:** MongoDB
- **Cloud Database:** MongoDB Atlas
- **Data Model:** Document-based NoSQL storage
- **Backend Integration:** Database models and schemas
- **Configuration:** Database connection information maintained through environment variables

6.7.2 Main Data Models

1. **User Model**
   Purpose: Stores information of the users.
   Main Information:

- User name or organization name
- Email
- Password
- Contact information
- Location
- User role
- NGO verification-related status where applicable

2. **Donation Model**
   Purpose: Stores information regarding food donation.
   Main Information:
   - Donor reference
   - Food type
   - Quantity
   - Expiry information
   - Pickup time
   - Pickup location
   - Donation status

3. **Donation Request Model**
   Purpose: Information of request of donation made by NGO is stored here
   Main Information:
   - Donation reference
   - NGO reference
   - Request status
   - Request-related information

4. **Notification Model**
   Purpose: To store notification
   Main Information:
   - User reference
   - Notification message
   - Notification information
   - Date and status information

5. **NGO Verification Model**
   Purpose: Information regarding Ngo verification is stored here to check NGO should be allowed in system or not.
   Main Information:
   - NGO reference
   - Verification document information
   - Verification status
   - Admin verification decision

5. Software Test Document (STD)

5.1 Introduction

Software Test Document Chapter ensures that all the functional and non-functional requirements and properly implemented in the system. It tests the defects in the system and gaps that need to be covered to make functional and non-functional requirements to be properly implemented. It tests the defects, verifies the correctness.

5.1.1 System Overview

Local Food Distribution system is a web based application to donate excess food and receive the donation. It has three main actors donor, NGO and admin. Donor donates the food, NGO accepts donations and admin manage them.

5.1.2 Test Approach

For the project the testing approach used is functionality and scenario based, it tests the authentication, all the view operations are working properly, with profile management tests and donor and NGO and admin based interaction tests.

5.2 Test Plan

5.2.1 Features to be Implemented

1. Authentication
2. View Operations
3. User Signup
4. Donor Management System
5. Profile View and Update
6. NGO Donation Interaction Lifecycle
7. Admin User and NGO Verification Management
8. Donation Approval and Status Control

5.2.2 Test Cases

TC-1: Authentication

Table 4.1 represents the test case of User Authentication Test Case, it makes sure that system is connected to data base, there is use in system and user is trying to login in, then it validates the authentication.

ID	1

Name	User Authentication Test Case
Explanation	This test case verifies the authentication mechanism for registered users that include Donor, Admin and NGO.
Setup	1. The system is connected to database and is running. 2. At least one user exists in the system. 3. User is on the login screen.
Expected Outcome	1. User is authenticated. 2. User is redirected to respective module. 3. On logout, User is redirected back to login screen
Verdict Pass/Fail	Pass

Table 5.1: Authentication

TC-2: View Operations

Table 4.2 represents the test case of All view-only operations Test Case, it makes sure that all the view only operations in the system that does not involve the modifying part in it are correctly verified.

ID	2
Name	All view-only operations
Explanation	This test case verifies that all the view only operations are working successfully and displaying respective outcomes. It checks that all the notifications can be viewed, Donor and NGO can view their donations history, Ngo can view available donations and admin can view all the donations.
Setup	1. User has logged in. 2. Respective data in accordance with request is displayed on the system.
Expected Outcome	1. List and details are displayed. 2. Data is shown according to user role. 3. Filter works accordingly. 4. Appropriate message is shown in case there is no data.
Verdict Pass/Fail	Pass

Table 5.2: View Operations

TC-3: User Signup

Table 4.3 represents the test case of User Registration Test Case, it makes sure that all the registration flows are working finely and required validations are enforced correctly and correct status is assigned to the user.

ID	3
Name	User Registration Test Case
Explanation	This test case verifies the functionality for donors and NGOs signup.
Setup	1. User is on signup screen.

	2. System is connected to database.
Expected Outcome	1. If donor has signed up then its account is successfully created. 2. If NGO has signed up that its account is created with unverified status. 3. Duplicate registrations are rejected.
Verdict Pass/Fail	Pass

Table 5.3: User Signup

TC-4: Donor Donation Management

Table 4.4 represents the test case of Donation Management Test Case; it makes sure that all the donation operations on the side of donor are working properly and gives the test case setup, expected outcome and verdict pass/fail.

ID	4
Name	Donation Management Test Case
Explanation	This test case verifies the donor side donation operation posting, editing and deleting donations are working finely.
Setup	1. Donor is logged in. 2. Donations exist for operation edit and delete.
Expected Outcome	1. Donation is posted with available status. 2. Donation details are updated successfully when updating available donation. 3. When donation with available status is deleted it is successfully removed from database.
Verdict Pass/Fail	Pass

Table 5.4: Donor Donation Management

TC-5: Profile View and Update

Table 4.5 represents the test case of Profile Management Test Case, it makes sure that when the profile of the donor or NGO is viewed and updated it performs this operation successfully and gives the test case setup, expected outcome and verdict pass/fail.

ID	5
Name	Profile Management Test Case
Explanation	This test case verifies the viewing and editing of user profiles.
Setup	1. User has logged in. 2. Profile data exists.
Expected Outcome	1. Profile information is displayed successfully. 2. Editable fields of the user are updated successfully. 3. Non editable fields cannot be edited.
Verdict Pass/Fail	Pass

Table 5.5: Profile View and Update

TC-6: NGO Donation Interaction Lifecycle

Table 4.6 represents the test case of NGO Donation Interaction Test Case, it verifies all the interactions of NGO with donation and checks their expected outcomes.

ID	6
Name	NGO interaction lifecycle
Explanation	This test case verifies the expected outcomes of NGO requesting available donation and marking booked donations as collected.
Setup	1. NGO is logged in. 2. There is at least one donation in the system.
Expected Outcome	1. On requesting donation, donation status is changed to requested. 2. On requesting donation , a donation request is created with pending status. 3. Booked donation can be marked as collected and donation status is changed to collected.
Verdict Pass/Fail	Pass

Table 5.6: NGO Interaction Lifecycle

TC-7: Admin User and NGO Verification Management

Table 4.7 represents the test case of Administrative Management Test Case, it verifies all the admin operations related to user management and NGO verification

ID	7
Name	Admin user and NGO verification management
Explanation	This test case verifies the expected outcomes of Admin viewing and deleting users and verifying the NGO
Setup	1. Admin has logged in. 2. At least one NGO request exists. 3. At least one user is in the system.
Expected Outcome	1. On approving NGO, NGO status updates to verified or unverified. 2. When user is deleted by admin, User is removed safely. 3. Respective notifications are sent to users accordingly.
Verdict Pass/Fail	Pass

Table 5.7: Admin User and NGO Verification Management

TC-8: Donation Approval and Status Control

Table 4.8 represents the test case of Donation Approval and Status Management Test Case, it verifies the admin's control on the donation requests.

ID	8
Name	Admin donation approval and status control
Explanation	This test case verifies the expected outcomes of Admin accepting the requested donations of NGOs and managing donation status.
Setup	1. Admin is logged in. 2. At least one donation request exists for donation. 3. At least one donation exists in the system.
Expected Outcome	1. One NGO request for each donation is accepted. 2. On accepting one NGO's request, all other NGO's requests for that donation are auto rejected. 3. Donation status is changed to booked. 4. On admin changing status from one status to another, that status is updated. 5. Invalid status changes are not updated.
Verdict Pass/Fail	Pass

Table 5.8: Donation Approval and Status Control

6. Conclusion

6.1 Conclusion

In conclusion, we implemented a system which connects the donor and NGO under proper monitoring and a system that displays proper transparency of donation.

It implements authentication, authorization, donation management, NGO requests management, documents verification and admin, NGO and donor workflow properly and accurately.

The use of React.js, node.js, express.js, and MongoDB provided scalable and maintainable architecture for the project. Authentication, validation, authorization and error handling improved the reliability and security of the system.

Overall, the project helps in reducing the wastage of surplus food and improves food distribution effectively and efficiently.

6.2 Future Work

There are a lot of improvements and work that can be done on the system to make it more reliable for users and make it more user friendly.

Mobile Application development:

It can be made as a mobile app so that it is easier for users to access and use it. The user would then not have to open browser and login every time to donate, and they can install the app to access it.

Real Time Chat:

Allowing donor and NGO to communicate directly would be more effective. It will make it a lot easier for both Ngo and donor when the donation is approved for the NGO, then that NGO can directly message the donor and communicate and ask questions regarding it.

Location-Based Matching:

Match donations with nearby NGOs using geolocation services. It will allow the NGO to view the food donations that are nearby and it will make it a lot easier for NGO to take decision that whether they want this donation or not and will they be able to reach destination on time before the food expires.

External Integration:

The system could be connecting food banks and charities to launch this project on greater scale so the mor amount of food is donated and more NGO are then interested in it.

REFERENCES

1. Delaney, D., & Brown, S. (2002). Document templates for student projects in software engineering (Technical Report NUIM-CS-TR2002-05). Department of Computer Science, National University of Ireland, Maynooth. https://www.academia.edu/14412345/DOCUMENT_TEMPLATES_FOR_STUDENT_PROJECTS_IN_SOFTWARE_ENGINEERING
2. Institute of Electrical and Electronics Engineers. (1998). IEEE recommended practice for software requirements specifications (IEEE Std 830-1998). http://www.math.uaa.alaska.edu/~afkjm/cs401/IEEE830.pdf
3. [1] TIMESOFINDIA.COM. (2019, September 17). Fight hidden hunger with wheat. The Times of India. https://timesofindia.indiatimes.com/life-style/food-news/fight-hidden-hunger-with-wheat/articleshow/71148423.cms
4. [2] Vavista. (n.d.). 7 ways you can reduce your food waste. Vavista. https://vavista.com/blog/7-ways-you-can-reduce-your-food-waste/
5. [3] Eby, K. (2024, September 30). Complete guide to agile workflow diagrams with examples and templates. Smartsheet. https://www.smartsheet.com/content/agile-project-management-workflow
6. Live project: www.foodbridge.tech
7. GitHub Link: https://github.com/sehrishsiddique853/local-food-charity-system.git